

LainePoiss[®]—A Lightweight and Ice-Resistant Wave Buoy

VICTOR ALARI,^a JAN-VICTOR BJÖRKQVIST,^{b,a,c} VALDUR KALDVEE,^d KRISTJAN MÖLDER,^d SANDER RIKKA,^a ANNE KASK-KORB,^e KAIMO VAHTER,^a SIIM PÄRT,^a NIKON VIDJAEV,^a AND HANNES TÖNISSON^f

^a Department of Marine Systems, Tallinn University of Technology, Tallinn, Estonia

^b Norwegian Meteorological Institute, Bergen, Norway

^c Marine Research, Finnish Meteorological Institute, Helsinki, Finland

^d WiseParker OÜ, Tallinn, Estonia

^e Estonian Maritime Academy, Tallinn University of Technology, Tallinn, Estonia

^f Institute of Ecology, Tallinn University, Tallinn, Estonia

(Manuscript received 1 July 2021, in final form 4 December 2021)

ABSTRACT: Wave buoys are a popular choice for measuring sea surface waves, and there is also an increasing interest for wave information from ice-covered water bodies. Such measurements require cost-effective, easily deployable, and robust devices. We have developed LainePoiss (LP)—an ice-resistant and lightweight wave buoy. It calculates the surface elevation by double integrating the data from the inertial sensors of the microelectromechanical system (MEMS), and transmits wave parameters and spectra in real time over cellular or satellite networks. LP was validated through 1) sensor tests, 2) wave tank experiments, 3) a field validation against a Directional Waverider, 4) an intercomparison of several buoys in the field, and 5) field measurements in the Baltic Sea marginal ice zone. These extensive field and laboratory tests confirmed that LP performed well (e.g., the bias of H_{m0} in the field was 0.01 m, with a correlation of 0.99 and a scatter index of 8%; the mean absolute deviation of mean wave direction was 7°). LP was also deployed with an unmanned aerial vehicle and we present our experience of such operations. One issue that requires further development is the presence of low-frequency artifacts caused by the dynamic noise of the gyroscope. For now, a correction method is presented to deal with the noise.

SIGNIFICANCE STATEMENT: Operational wave buoys are large and therefore expensive and inconvenient to deploy. Many commercially available devices cannot measure short waves and are not tested in ice. Our purpose was to develop an affordable wave buoy that is lightweight, ice resistant, capable of measuring short waves, and also has a longer operating life than existing research buoys. The buoy is easily deployed with a small boat or even an industrial drone, thus reducing operating costs. The buoy is accurate, and captures waves that are too short for operational wave buoys. This is relevant for coastal planning in, e.g., archipelagos and narrow fjords. We measured waves in ice in the Baltic Sea, and are planning to extend these measurements to Antarctica.

KEYWORDS: Sea ice; Wind waves; Air-sea interaction; Buoy observations

1. Introduction

Wind-generated waves with periods of up to 25 s dominate the spectrum of ocean surface vertical variance (Munk 1950; Holthuijsen 2007). These waves are observed visually, but an objective quantification of wave heights, lengths, and propagation directions requires measurements with in situ or remote sensing technologies. This paper focuses on a newly developed wave measuring buoy.

In situ instruments do not measure wave properties directly. At the sea surface, wave buoys should follow the three-dimensional movement of water particles while measuring the inertial data or the Doppler shift of a GPS signal (Herbers et al. 2012). Below the sea surface, pressure transducers measure wave-induced pressure fluctuations (Cavaleri

1980), Doppler current meters measure the wave-induced orbital motions (Gordon and Lohrmann 2002), and inverted echo sounders measure acoustic travel time between the device and the sea surface (Wadhams 1978). Wave gauges piercing the sea surface measure the up-and-down movement of water through electrical capacitance or resistance (Donelan et al. 1985; Graber et al. 2000). The surface wave spectra is then calculated from the measured physical quantity by mathematical transformations and wave theory.

All the measurements techniques in the above (nonexhaustive) list are feasible, but have their own limitations and peculiarities, for example, gaps in data due to salty water over washing the buoy and blocking the GPS signal (Björkqvist et al. 2016), spurious data in echo soundings generated by breaking waves, and diminishing of short waves in pressure recordings (Bishop and Donelan 1987). More generally, the size of the instrument, the basic measurement principle, and the meteorology–ocean conditions affect the measurement result. Besides inherent limitations, operational and practical considerations, like ease of use and cost of the instrument, might be important.

 Denotes content that is immediately available upon publication as open access.

Corresponding author: Victor Alari, victor.alari@taltech.ee

DOI: 10.1175/JTECH-D-21-0091.1

© 2022 American Meteorological Society. For information regarding reuse of this content and general copyright information, consult the AMS Copyright Policy (www.ametsoc.org/PUBSReuseLicenses).

Brought to you by University of Maryland, McKeldin Library | Unauthenticated | Downloaded 01/07/24 05:26 PM UTC

Majority of operational wave measurements are done with surface-following buoys, which may also include sensors for recording atmospheric and upper-ocean state variables (Thomson 2012). These buoys can be tethered to the sea floor or allowed to drift under the influence of wind and currents. An advantage of these buoys is their long operational life (several years). However, the established operational buoys are large and heavy. The National Data Buoy Center of NOAA has tailored meteorology–ocean buoys, equipped with wave measuring sensors, ranging from 1.8 to 12 m in diameter. The widely used Datawell Waverider Mk-III, TRIAXYS Directional Wave Buoy, and Fugro Seawatch Mini II Buoy have diameters of at least 70 cm and weights of over 100 kg. These off-the-shelf operational buoys may not be an affordable choice for simultaneous deployments, and their handling requires expertise as well as vessel for deployment. These issues are somewhat alleviated with buoys as the Datawell DWR-G4, which weighs 17 kg, has a diameter of 40 cm, and has a battery life of 30 days.

Recent advances in accurate and low-cost motion sensors and GPS technology have led to the development of a low-cost easy-to-handle wave measurement platform called Spotter (Raghukumar et al. 2019; Lancaster et al. 2021). It is a sea state detector with unprecedented coverage (Smit et al. 2021; Houghton et al. 2021) and thanks to its small size, it might turn out to be a practical tool for estimating wind speed from wave spectra (Voermans et al. 2020). First field experiments of measuring waves in grease ice with Spotter show promising results in performing these types of observations at a low cost (Kodaira et al. 2020). The simultaneous deployments of several (tens, hundreds) drifting wave buoys is also beneficial for understanding wave–current interaction at coastal and oceanic scales (Pearman et al. 2014; Veras Guimarães et al. 2018) and wave growth in complex archipelagos (Björkqvist et al. 2019) and fjords (Christakos et al. 2021).

In this paper, we describe and validate a new wave buoy called LainePoiss (LP), which uses the microelectromechanical system (MEMS) inertial measurement unit to detect surface motion. LP was originally designed for wave measurements in ice, but has over 3 years of research and development matured into a valid device; it is already used for real-time wave monitoring in the Baltic Sea. When developing the buoy, we have kept the following combination of performance characteristics in focus: ice resistant, lightweight, operational, small, and affordable. These characteristics allow LP to be used for various research and engineering applications. For example, we are currently planning to use LP for wave measurements in the marginal ice zone, extending measurement times of operational wave buoys in the seasonally ice covered Baltic Sea, field measurements of shorter waves in archipelagos and lakes, and using unmanned aerial vehicles (UAV) as a rapid method for deployments. LP is also a core infrastructure for validating operational coastal wave models in Estonia.

The structure of the paper is as follows. In section 2, we describe the wave buoy in detail and introduce algorithms for converting the acceleration data into displacement data. In section 3, the omnidirectional and directional wave parameters are defined. Section 4 describes different laboratory tests

for sensor validation, including standstill measurements of acceleration noise and benchmark tests with monochromatic motions. Thereafter, we will describe the accuracy of the buoy in capturing high-frequency waves in a wave tank. Section 5 deals with the field test results. This includes an extensive validation campaign against a Directional Waverider and an intercomparison of several LP buoys. We also describe drifting experiments results, where a UAV was used as a deployment method. We end section 5 by describing the results of waves-in-ice measurements from the Baltic Sea. In section 6, we discuss the limits and merits of the new technology and conclude our main findings in section 7. Four appendixes end the paper, with the description of the inertial sensor, description of the denoising procedure, calculating validation statistics, and describing our experience of UAV deployments.

2. Materials and methods

a. Technical description of the buoy

LP is a spherical wave buoy with a diameter of 32 cm, a height of 22 cm (Fig. 1), and a weight of 3.5 kg. The enclosure is made of two tough glass fiber halves. Between the halves, there is a seal that makes the hull waterproof after it has been bolted together. After the buoy is sealed, it can be turned on and off by holding a magnet to the outside of the hull. Rechargeable lithium-ion batteries, with a capacity of 335 W h, can power the buoy for roughly 2 months.

The microcontroller of the buoy is connected to the sensors, the memory, and the communication modems through a custom built PCB board (Fig. 2). This controller—using an ARM Cortex-M4 core—was chosen because of its low power consumption, floating point support, and digital signal processing functionality.

The inertial sensors of the buoy are a 3D accelerometer, a 3D gyroscope, and a magnetometer. These sensors are part of the Xsens MTi-3 (see appendix A for more details) attitude and heading reference system (AHRS). The internal processor of the system synchronizes the sensors, applies calibration models, and runs the sensor fusion algorithm (including the extended Kalman filter), in order to convert inertial data from buoy body reference frame to Earth reference frame. Henceforth, raw data will mean the original acceleration data that are transformed to the Earth reference frame by the sensor. After assembling the buoy, we performed a magnetic calibration that corrected for the disturbance inflicted to the magnetic field of Earth by ferromagnetic materials. The device location was logged by a Global Navigation Satellite System (GNSS) receiver.

We configured the output frequency of the inertial data to 50 Hz, and these data are written to an SD card. Depending on the configuration set by the user, the raw 50 Hz data and the processed data (wave spectra and bulk parameters) are also sent to a cloud server in real time (e.g., every 30 min) using cellular LTE or only processed data over satellite Iridium SBD networks. The buoy can be configured to use

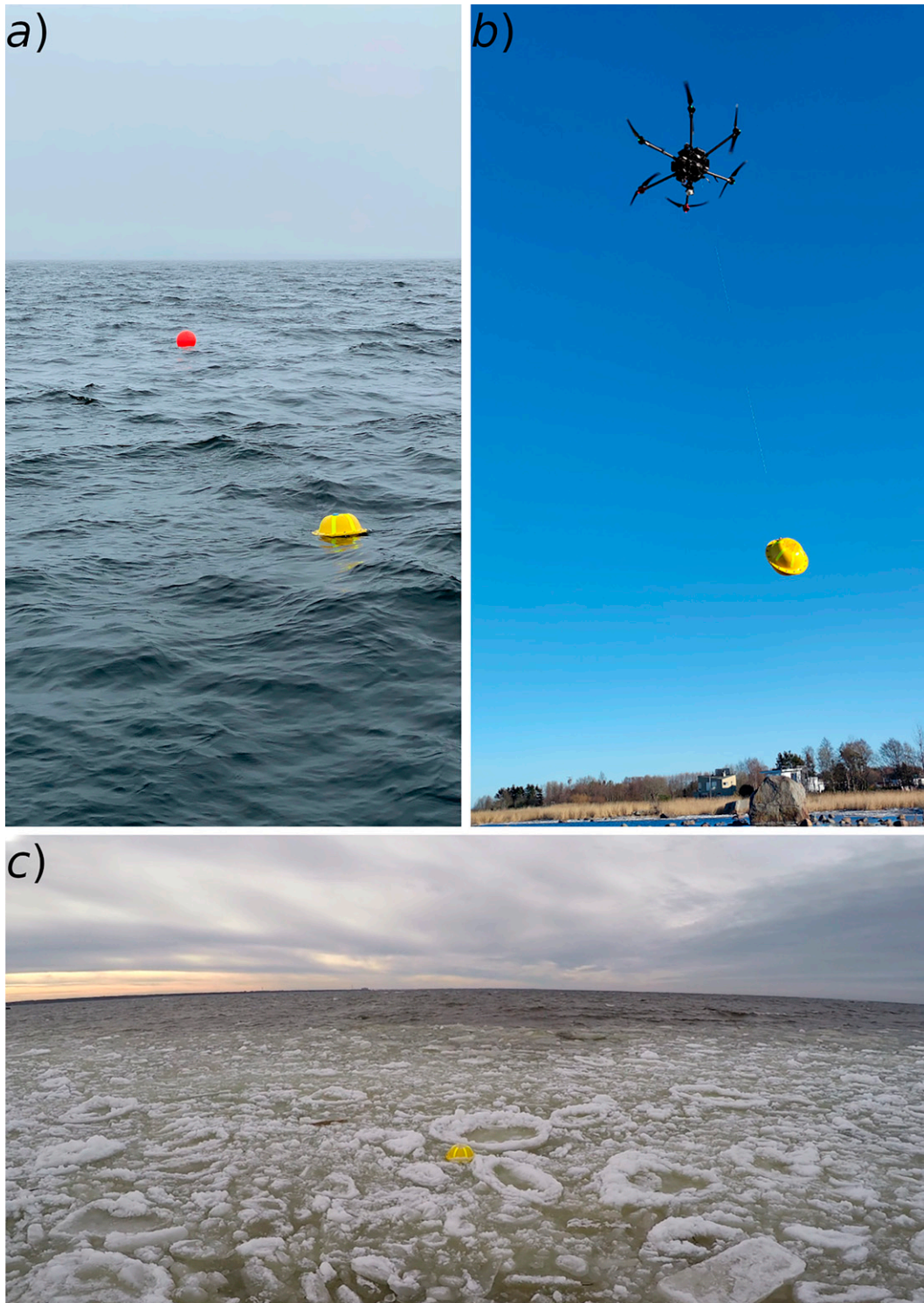


FIG. 1. (a) LainePoiss in a moored configuration, (b) transporting with a UAV, and (c) drifted to pancake ice.

either one or both of these communication options; the buoy always prioritizes the cellular network, but can automatically switch to the satellite network if no cellular coverage exists. We chose the cellular network as the primary communication

channel because of its significantly lower data transmission costs. The buoy can therefore also send the raw data through the cellular network, although this option can be disabled to save power.

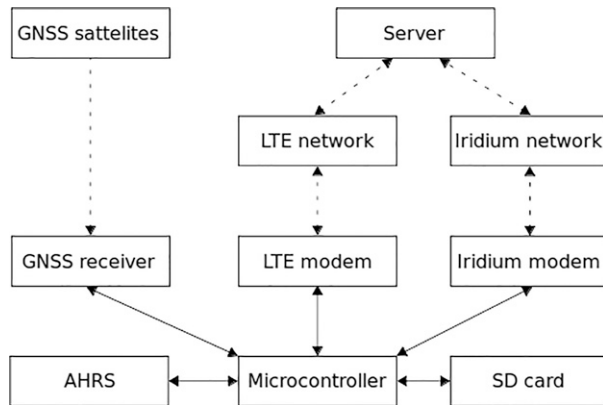


FIG. 2. Scheme of the electronics components of the buoy.

b. Data processing

The buoy sampled the acceleration with a 50 Hz sampling frequency, and (in the field tests) these data were saved in files containing 65 538 measurements (circa 22 min). We have used the raw data which are already transformed into an Earth reference frame by the sensors fusion algorithms. For each deployment, these acceleration data were combined into one long time series from which we constructed 30 min displacement time series with starting times of :00 and :30 using the following procedure:

- 1) We constructed 32 min time series from the continuous acceleration data (:59–:31).
- 2) The x , y , and z accelerations were low-pass filtered (filter “biting” between 1.28 and 5.12 Hz) with a finite impulse response (FIR) filter that had 162 coefficients before being downsampled to a 5.12 Hz resolution using a nearest neighbor interpolation. Because of the original 50 Hz sampling frequency, this resulted in a maximum shift on 100 ms of the measurement.
- 3) The acceleration data were double integrated in Fourier space. Before integration, the white noise was removed for frequencies up to 0.10 Hz, and the amplitudes below 0.05 Hz were set to zero (these cutoffs were appropriate for our specific dataset; see [appendix B](#) for details). We removed 30 s from the start and end of the double integrated time series to cut out transient data caused by the Fourier integration.
- 4) Finally, after accounting for the data lost by FIR filtering and Fourier integration, we cut the time series to start exactly at :00 or :30 to a 9216 point long block. These 30 min block coincided exactly with the displacement time series of the Waverider at Suomenlinna, which we used for validation.

The denoising of the acceleration data was performed by estimating the mean amplitude of the fast Fourier transform (FFT) of the signal between frequencies 1/100 and 1/30 Hz. The integration was performed in Fourier space, where the removing of the low-frequency energy was applied gradually below 0.06 Hz using a half-cosine function, setting the

amplitudes to 0 below 0.05 Hz. This integration procedure is similar to the one used in [Rabault et al. \(2020\)](#). For the field data, a correction to the energy was performed based on the amount of variance that was lost in the denoising. The denoising, compensation and integration procedures are documented in detail in [appendix B](#).

The 30-min displacement time series were then used to calculate wave spectra from 100-s blocks using the Welch method with a 50% overlap (35 segments), resulting in a wave spectrum with a 0.01 Hz resolution. Each segment was tapered with a Hann window. The highest frequency used in the spectrum was 1.28 Hz, which corresponds to a 95 cm deep water wavelength and approximately 3 times the buoy diameter. Higher frequencies were starting to be distorted by the size of the buoy when compared to the wave gauge data (see [section 4c](#) on the wave tank experiment).

Note that the data processing procedures in the laboratory experiments differed slightly from the data collected from field tests, since the time series were shorter. Please see [section 4c](#) for details on the exact procedure for laboratory data. For the waves-in-ice measurements ([section 5d](#)), the data processing procedure differed also a bit from the other field data—namely, the energy after denoising was not added back because the noise was so dominant.

3. Wave parameters

a. Omnidirectional parameters

We used the following definitions of wave parameters. The spectral moments are defined as

$$m_n = \int_{f_0}^{f_1} f^n E(f) df, \quad (1)$$

where a lower integration frequency of $f_0 = 0.10$ Hz and an upper frequency of $f_1 = 0.58$ Hz was used to match the highest frequency that was reliably captured by the Waverider.

Using the spectral moments we defined the significant wave height:

$$H_{m_0} = 4\sqrt{m_0}. \quad (2)$$

The wave periods were defined as

$$T_{m_{-10}} = \frac{m_{-1}}{m_0}, \quad (3)$$

$$T_{m_{01}} = \frac{m_0}{m_1}, \quad (4)$$

$$T_{m_{02}} = \sqrt{\frac{m_0}{m_2}}, \quad (5)$$

$$T_p = [\arg\max E(f)]^{-1}, \quad (6)$$

$$T_c = \frac{\int_{f_0}^{f_1} f^{-1} E(f)^4 df}{\int_{f_0}^{f_1} E(f)^4 df}. \quad (7)$$

Here, the peak period T_p was determined using a parabolic fit. The so-called characteristic period T_c is of the same type that was originally developed as an alternative to peak frequency by Young (1995). Björkqvist et al. (2019) proposed it as a more stable alternative characterization of a representative wave frequency in archipelago conditions, where the peak period is often ill defined. Note that we use a slight modification by integrating a weighted average of the inverse of the frequency (f^{-1}) compared to the weighted integration of f by Young (1995) and Björkqvist et al. (2019).

The narrowness of the spectrum was determined by the κ^2 -narrowness parameter (Battjes and van Vledder 1984):

$$\kappa^2 = \frac{1}{m_0^2} \left\{ \left[\int_{f_0}^{f_1} E(f) \cos(2\pi f T_{m_{02}}) df \right]^2 + \left[\int_{f_0}^{f_1} E(f) \sin(2\pi f T_{m_{02}}) df \right]^2 \right\}, \quad (8)$$

where κ^2 takes values between 1 (infinitely narrow spectrum) and 0 (white noise) and has been found to capture the spectral shape better (especially in the archipelago) than width parameters depending on high moments (m_2 or m_4) (Björkqvist et al. 2019).

b. Directional parameters

Directional parameters were calculated using the first pair of Fourier coefficients, $a_1(f)$ and $b_1(f)$. These coefficients were calculated from the cross-spectra following Longuet-Higgins (1961):

$$a_1(f) = \frac{Q_w(f)}{\sqrt{C_w(f)[C_{nn}(f) + C_{ee}(f)]}}, \quad (9)$$

$$b_1(f) = \frac{Q_w(f)}{\sqrt{C_w(f)[C_{nn}(f) + C_{ee}(f)]}}, \quad (10)$$

where $Q(f)$ and $C(f)$ are the quadrature- and cospectra, with subscripts v , e , and n referring to the vertical, east, and north displacements.

The mean direction at each frequency was calculated as

$$\theta_m(f) = \arctan \left[\frac{a_1(f)}{b_1(f)} \right] + 180^\circ, \quad (11)$$

with the mean and peak direction being

$$\theta_m = \arctan \left[\frac{a_1}{b_1} \right] + 180^\circ, \quad (12)$$

$$\theta_p = \theta(f_p) = \arctan \left[\frac{a_1(f_p)}{b_1(f_p)} \right] + 180^\circ, \quad (13)$$

where f_p is the peak frequency determined without a parabolic fit.

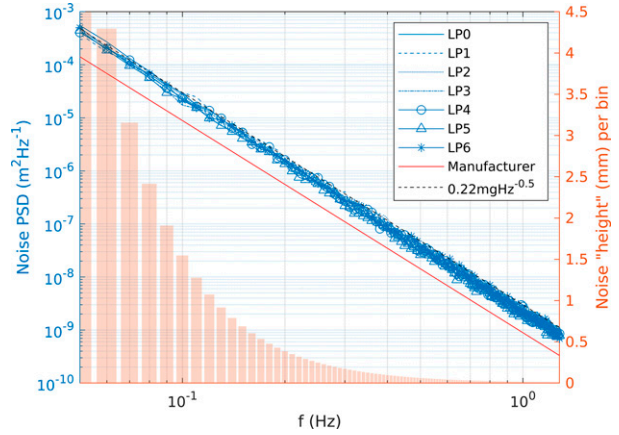


FIG. 3. Noise level of a new (LP0) and used (LP1-LP6) AHRS in comparison with manufacturers value. The noise height is calculated using the fit value to the real sensors.

The directional spread was defined as

$$\sigma(f) = \sqrt{2 - 2m_1^F(f)}, \quad (14)$$

where

$$m_1^F(f) = \sqrt{a_1^2(f) + b_1^2(f)}. \quad (15)$$

The mean and peak spreads were defined as for the directional spread [Eq. (14)] using

$$\bar{m}_1^F = \sqrt{\bar{a}_1^2 + \bar{b}_1^2}, \quad (16)$$

$$m_1^F(f_p) = \sqrt{a_1^2(f_p) + b_1^2(f_p)}. \quad (17)$$

4. Laboratory tests

a. Sensor static noise

The manufacturer of the MEMS sensor reports a static noise density value of $0.12 \text{ mg}(\sqrt{\text{Hz}})^{-1}$ for the acceleration sensor (mg is milli g ; it has a value of $9.81/1000$). To test if this represents the actual noise of a single sensor, we conducted a standstill measurement, where, at a room temperature of 21°C , we let the sensor measure for 3 h. We then calculated the power spectrum of the acceleration noise and transformed it to noise displacement spectra by dividing the acceleration spectra with $(2\pi f)^{-4}$. We found that the unused sensor has a higher noise density than the manufacturer's value (Fig. 3). We repeated the same procedure for sensors already used in deployments and found that the noise does not increase with the usage of the sensor (LP1 was used the most, 5100 h). Therefore, for the sensor noise, we will use the value $0.22 \text{ mg}(\sqrt{\text{Hz}})^{-1}$. From a practical point of view, this sensor noise level only becomes important when dealing with very low amplitude waves, e.g., waves in ice. For the field measurements described in this paper, the static noise

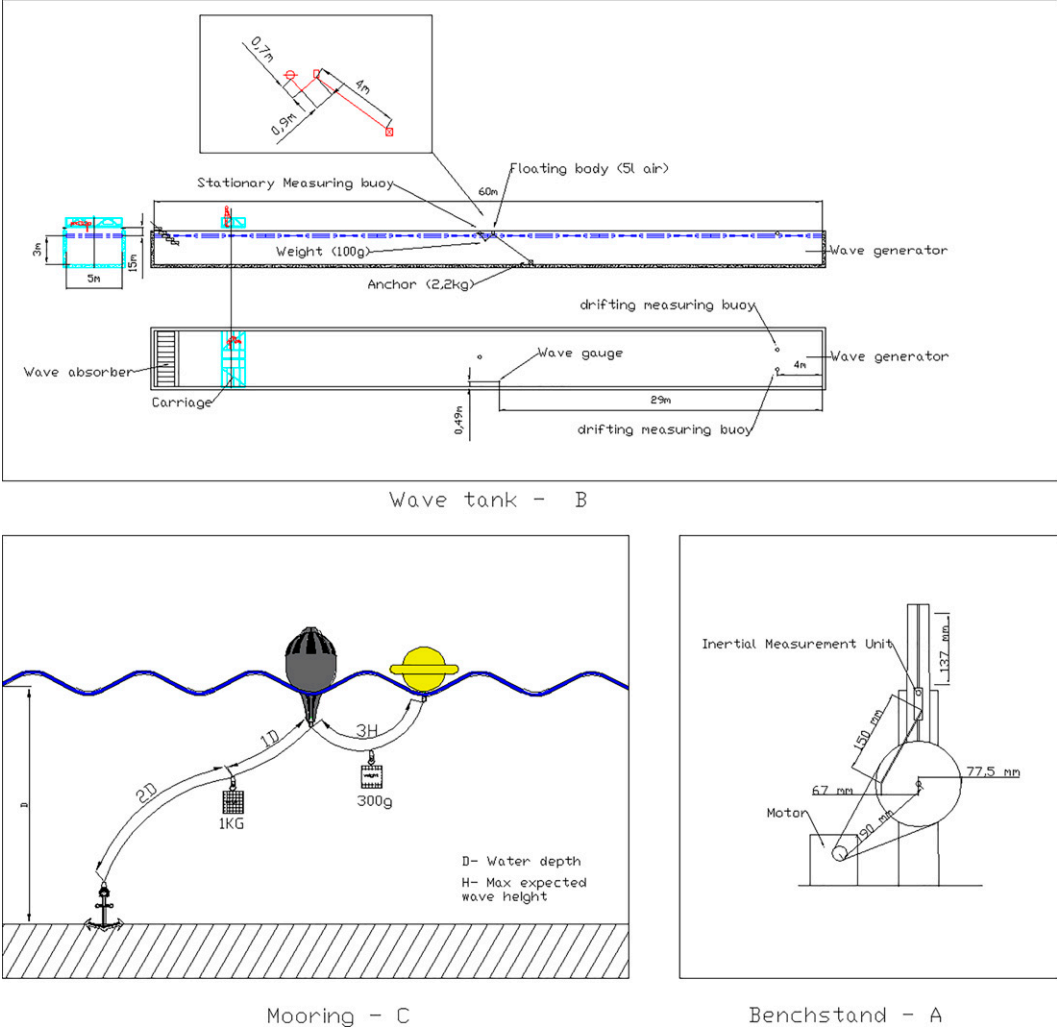


FIG. 4. Technical drawings of (a) benchstand, (b) wave tank, and (c) recommended mooring setup.

of the sensor was subtracted only for the waves-in-ice measurements (section 5d).

b. Benchmark test

The benchmark tests were conducted with a custom-made device (Fig. 4a). The amplitude was kept constant at 68 mm (± 1 mm), while period was varied between 1 and 25 s by adjusting the speed of the device. Since the device did not reproduce a perfect sine wave, all the measured acceleration signals had harmonics at multiples of the dominant signal. The measured acceleration signal was transferred to frequency space, the noise was removed and then double integrated and transferred back to time space. Mean amplitudes of N cycles were calculated. N varied between 8 and 30, depending on the cycle period.

The results of the benchmark test for periods 1, 5, 10, 20, and 25 s are shown in Table 1. The sensor is able to measure movement in the wind-wave and swell period range. The

difference between the prescribed and measured value depends on the acceleration—the measured amplitude starts to deviate with decreasing acceleration, being 6% at 0.04 Hz.

c. Wave tank

We tested the wave buoy’s response to irregular waves in both tethered and free-drift setups. The experiments were conducted in a wave tank (Fig. 4b) owned by the Small Craft

TABLE 1. Test bench results.

Wave period (s)	Prescribed amplitude (mm)	Measured amplitude (mm)
1	68	69
5	68	66
10	68	64
20	68	64
25	68	64

TABLE 2. Wave tank results. H_{m0} was integrated between 0.30 and 1.28 Hz. The peak frequency (f_p) corresponds to the wave gauge.

f_p (Hz)	H_{m0} (m)			
	Wave gauge	LP1	LP3 (moored)	LP4
0.57	0.15	0.16	0.16	0.16
0.63	0.16	0.15	0.17	0.16
0.53	0.17	0.19	0.20	0.20
0.53	0.17	0.19	0.20	0.19
0.77	0.07	0.07	0.07	0.07
0.77	0.06	0.08	0.07	0.07
0.87	0.09	0.11	0.11	0.10
0.73	0.10	0.10	0.11	0.10

Competence Centre of the Tallinn University of Technology. The aim was to validate the high-frequency part of the spectrum and determine the accuracy of LP in conditions which represent wave growth at short fetches.

The 60-m-long, 5-m-wide, and 3-m-deep tank uses 6 paddles to generate waves, which are recorded by a capacitance wave height gauge (developed by Akamina Technologies) in the middle of the tank. The duration of wave generation varied from 210 to 300 s.

For the tests, we moored one buoy 2 m from the wave gauge and let the other two drift. With the JONSWAP spectrum, we made 8 tests (Table 2), in which each combination of peak period and significant wave height was repeated twice. For each test, a 3-min-long time series was analyzed. The time series was converted into a displacement power spectra using the Welch method (block length 30 s, 50% overlap). Neither datasets were decimated; LP data sampled at 50 Hz and wave gauge data sampled at 200 Hz were used.

Significant wave height, integrated between 0.30 and 1.28 Hz, shows a satisfactory match between wave gauge and wave buoys (Table 2). The difference in significant wave height between the buoys and the wave staff was -1 to 3 cm. The wave spectra (Fig. 5) reveals a similar structure between the wave gauge and buoys up to 1.28 Hz and a slight shift of frequencies of the drifting buoys due to Doppler shift.

5. Field tests

The field tests were conducted in the seasonally ice-covered Baltic Sea, which is an enclosed basin with a maximum fetch of 700 km. The deployments were made within 5 km from the coast (Fig. 6).

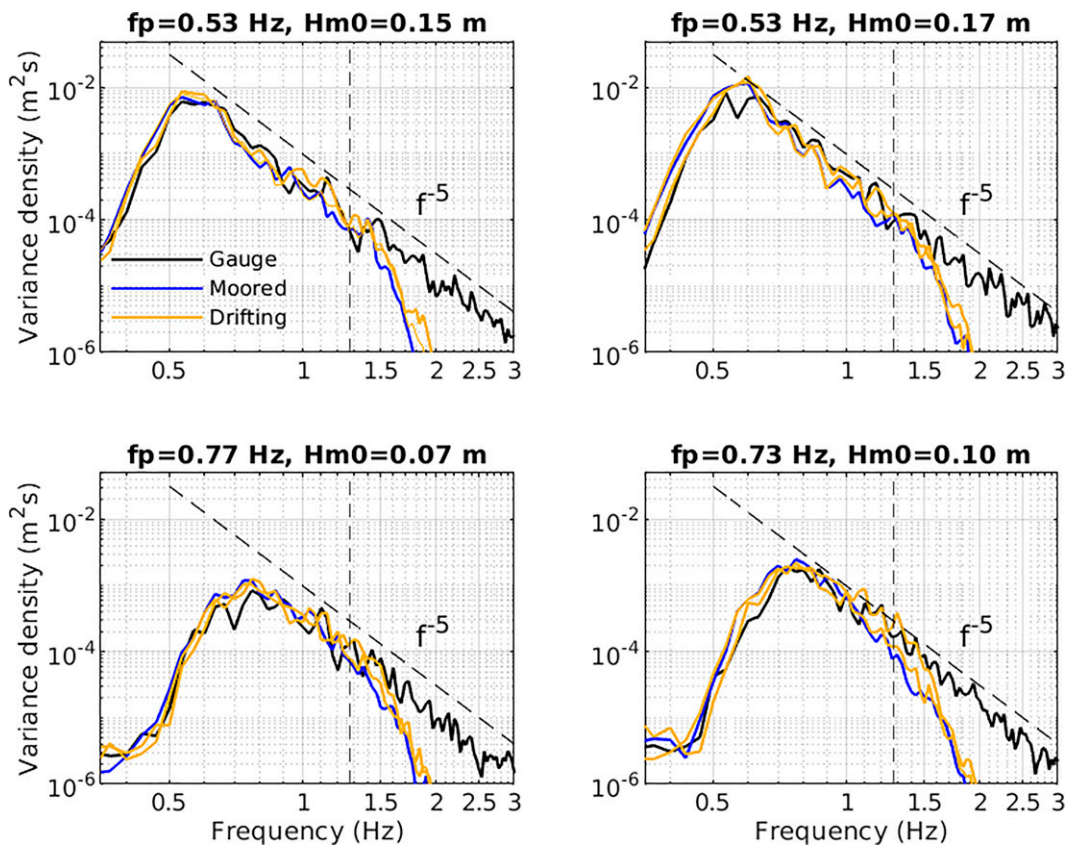


FIG. 5. Comparison of drifting and moored LP's with wave gauge at a wave tank. The wave field corresponds to the JONSWAP spectrum. The 1.28 Hz cutoff frequency is marked by a vertical dashed line.

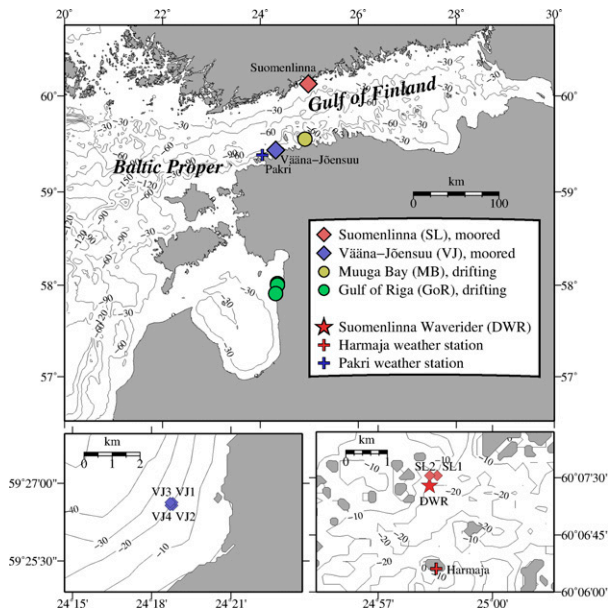


FIG. 6. Field test sites where LainePoiss's were moored (diamond) or deployed as drifters (circle). Existing operational wave and wind measurements are shown with a star and plus signs.

a. Comparison against a Directional Waverider

We deployed buoys at two locations close to the operational Directional Waverider DWR-Mk III,¹ which was moored at a depth of 20 m outside of Suomenlinna in the Finnish Archipelago (Fig. 6). The measurement area is characterized by tens of islands in varying size and busy recreational and commercial vessel traffic, although the commercial traffic during the deployments was reduced because of COVID-19 restrictions to travel.

With some short gaps, the deployments lasted from June to November 2020 (Table 3). Four different buoys were used, because the battery life of one device did not last for the entire campaign. The buoys were moored approximately 300 m north of the DWR (Fig. 6), where the water depth was 17 m. The recommended mooring consisted of an anchor, a surface float, LP, ropes connecting the assembly, and weights (Fig. 4c). The mooring endured strong winds from all major directions. The maximum 10 min averages were 15 m s^{-1} (west), 16 m s^{-1} (north), 14 m s^{-1} (east), and 21 m s^{-1} (south), with the gusts reaching 27 m s^{-1} . The buoy was not dragged from its deployment location, which shows that a 25 kg mooring anchor was sufficient.

Two buoys were deployed in June using a small plastic motorboat. In August, one of these buoys was removed and the second one was replaced (using the same boat). In September, the buoy was replaced again using a motorized sailing vessel, and it was finally retrieved with a pilot boat in the beginning of December.

¹ Then owned by the City of Helsinki, now owned by Finnish Meteorological Institute.

The spectra from both LP and DWR were integrated between 0.10 and 0.58 Hz (note that the actual low-frequency cutoff of the DWR spectra was 0.025 Hz), and both devices generally agreed on the wave statistics for the field experiment (Table 4; see appendix C for the definition of the statistical parameters). The mean significant wave height was 0.42 m, which is 0.10 m higher than the 2016–18 measured average (Björkqvist et al. 2019). Both LP and DWR measured a similar highest significant wave height, with the difference being within the expected sampling variability. The wave periods agreed better the more they were determined by the shorter waves, with T_{m02} showing the best agreement and T_p the worst. The maximum peak period values for LP are probably not realistic but stem from the low-frequency noise dominating the lowest integration frequency when total energy was very low. The dominant wave direction for both instruments was south-southwest. The spreading was, in a mean sense, slightly higher for LP than for DWR.

Overall, the significant wave height measured by LP and DWR matched well (Figs. 7–9), with a 0.01 m bias and 0.99 correlation coefficient (Table 5). During two occasions in the first deployment, LP measured an around 0.2 m higher significant wave height than the DWR (Fig. 7b). Both of these cases took place during around 10 m s^{-1} easterly winds, but LP and DWR matched up well during roughly 13 m s^{-1} easterly winds in the autumn (Fig. 9b). We have not yet determined a definitive reason for the discrepancy during the summer cases. A tangling of the moorings of the LPs is improbable since both LPs measured similar wave heights.

The wave periods determined from spectral moments compared well to those from the DWR (Figs. 10b,d,f). The scatter index was 3%–5%, with the biases being no more than 0.16 s (Table 5). The scatter index is highest (23%) for the peak period (Table 5). The general poor validation for the peak period is explained by the archipelago environments, where the peak of the spectrum is often not well defined (Björkqvist 2020). The wave spectrum was more closely unimodal for the strongest southerly winds, and the peak period therefore matched up well in the case of the highest significant wave heights (Fig. 10j). The characteristic period has been proposed as an alternative to the instances when the peak period is ill defined. The validation of the characteristic period showed that it removed a large part of the scatter between the two devices while keeping the bias at a low 0.03 s value (Fig. 10h, Table 5).

The wave direction measured by LP had a 1° – 2° bias with respect to the DWR (Table 5). While the agreement was good for both the mean and peak directions, the mean parameter expectedly has less scatter (Figs. 10c,e). A large scatter in directional peak parameters are expected (e.g., Pettersson et al. 2003), especially in the archipelago where the peak values are not even necessarily from the same wave component. For the same reason, also the peak directional spreading had a 25% scatter index even though the bias was only 2° (Table 5).

The mean spreading was more robust with a scatter index of only 9% and a correlation of 0.88. Still, LP systematically measured a slightly higher spreading than the DWR

TABLE 3. An overview of the field experiments where LainePoiss was moored.

	Deployment 1	Deployment 2	Deployment 3	Deployment 4
Device	LP1–LP2	LP3	LP4	LP1–LP4
Location	Suomenlinna (S1–S2)	Suomenlinna (S1)	Suomenlinna (S1)	Vääna-Jõesuu (V1–V4)
Depth (m)	17	17	17	19
Period	18 Jun–24 Jul 2020	14 Aug–21 Sep 2020	24 Sep–24 Nov 2020	22 Dec 2020–03 Jan 2021
Mean U_{10} (m s^{-1})	6.3	7.3	8.2	4.1
Mean H_{m0} (m)	0.32	0.39	0.50	0.51
Mean T_c (s)	4.5	4.4	4.3	5.8

(bias = 3° , slope = 1.09; Table 5). A visual comparison (Fig. 10g) reveals that the difference is better explained by a fixed offset than the calculated slope. Le Merle et al. (2021) found an offset of similar magnitude when comparing the (peak) directional spreading between radar and wave buoy measurements. The laboratory data of Lin et al. (2021) also showed that wave buoys can underestimate the directional spread compared to wave gauges with 10%. It is therefore possible that the difference between the spreading measured by LP and the DWR might partially be caused by the size difference between the devices. Lin et al. (2021) found that the impact of the mooring was small in the laboratory, but did not exclude that this factor would be more important in the field. All in all, the agreement in the directional parameters are good, especially considering the possible sources of uncertainty.

An additional validation of the spectral shape was performed by using the spectral narrowness parameter (κ^2). This parameter was proposed by Battjes and van Vledder (1984) and was found to be suitable for archipelago conditions by Björkqvist et al. (2019). The agreement between DWR and LP was reasonable (Fig. 10i), with a correlation of 0.92 and a bias of -0.02 (Table 5). The scatter index was high (23%), although we are not aware of any other cross-instrumental

TABLE 4. LainePoiss and DWR statistics during the deployments ($N = 6357$). N is smaller than the one used in the validation statistics Table 5 because in June–July 2020 two LPs were simultaneously measuring. In the validation, both buoys were included against the comparison with DWR; here only LP 1 is used (eastern buoy, Fig. 6).

Parameter	LP/WR			
	Mean	Std	Min	Max
H_{m0} (m)	0.42/0.42	0.27/0.29	0.04/0.03	1.84/1.90
T_{m02} (s)	3.05/2.91	0.40/0.43	2.11/1.91	4.87/4.53
T_{m01} (s)	3.26/3.06	0.46/0.49	2.17/1.93	5.48/4.84
T_{m10} (s)	3.84/3.44	0.61/0.59	2.44/2.05	6.73/5.86
T_c (s)	4.40/3.83	1.49/1.12	1.87/1.74	9.53/8.32
T_p (s)	4.72/3.99	2.10/1.41	1.72/1.72	10.00/8.54
κ^2 (–)	0.11/0.13	0.09/0.10	0.00/0.00	0.59/0.68
σ_m ($^\circ$)	45/41	12/11	27/23	80/80
σ_p ($^\circ$)	38/32	15/9	11/9	81/80
	Mode	Std	Min	Max
θ_m ($^\circ$)	202/198	45/44	5/1	359/359
θ_p ($^\circ$)	202/200	52/49	0/0	359/358

validation for this particular parameter. Several width parameters are defined using higher moments (Cartwright and Longuet-Higgins 1956) and are therefore sensitive to the high-frequency part of the spectrum. The validation was therefore partially limited by the 0.58 Hz upper frequency of the DWR. In the 0.10–0.58 Hz frequency range, LP spectra tend to be slightly wider than those from the DWR, which might be caused by low-frequency artifacts during low sea states.

The spectral comparison shows a qualitatively good match (Fig. 11) between LP and DWR, except for the low-frequency part where LP has artifacts. During the northerly wind case (Fig. 11a), the peak of the spectrum was roughly at 0.5 Hz, just below the cutoff frequency of the DWR. Above the DWR cutoff frequency, LP spectra followed an f^{-4} tail, which is in agreement with theory (Kitaigorodskii 1983; Phillips 1985). For the easterly and westerly wind events, a clear peak

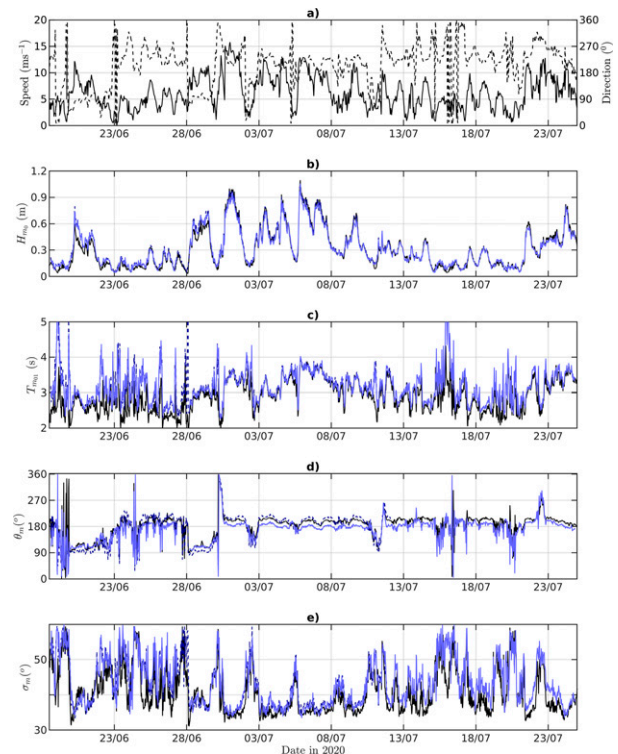


FIG. 7. Suomenlinna validation period 1. (a) Solid black line is wind speed and dashed black line is wind direction. (b)–(e) Black lines are DWR and blue lines are LP.

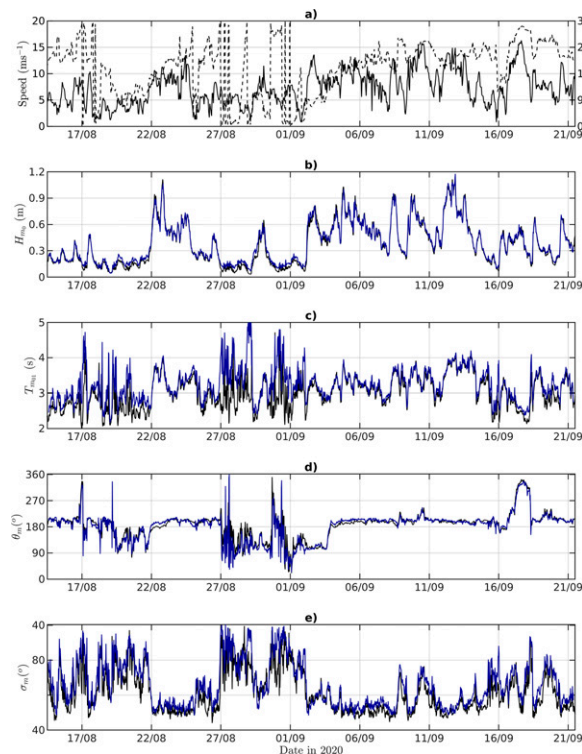


FIG. 8. As in Fig. 7, but for second validation period.

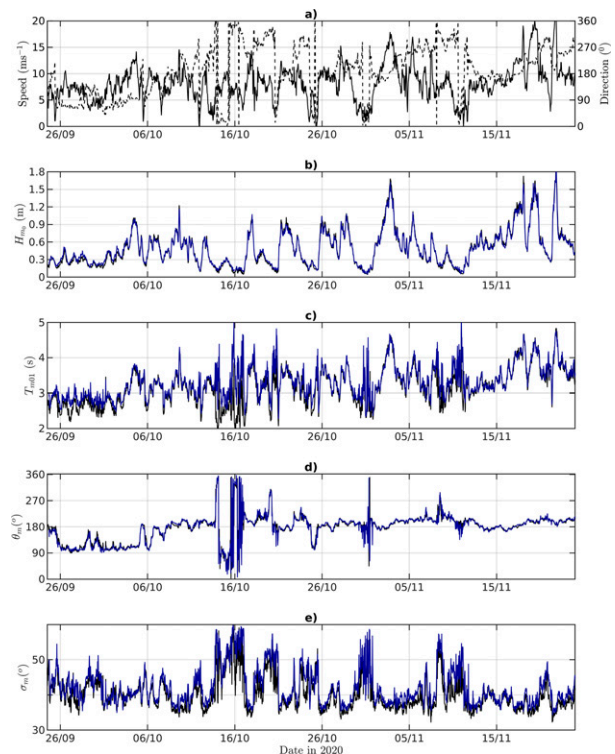


FIG. 9. As in Fig. 7, but for third validation period.

was not defined (Figs. 11b,c), which is a typical situation for the archipelago-type spectrum with many different fetches for a fixed wind direction (Björkqvist 2020). The highest waves measured during the campaign were due to southerly winds, and both the LP and the DWR spectra showed a good match even for the longer waves of a 9 s period (Fig. 11d).

In summary, the validation confirms the accuracy of LP in complex wave conditions. The scatter of the peak parameters is explained more by the ill-defined nature of the parameter than any actual differences between the devices. Otherwise, the validation statistics are similar to those determined for other small buoys when validated against established technologies (Raghukumar et al. 2019; Lancaster et al. 2021).

b. Intercomparison of four moored devices

For an intercomparison of several buoys in the same wave conditions, we moored four buoys in a square layout with a maximum distance of 140 m from each other in places where the water depth was 18–19 m (deduced from nautical charts) (Fig. 6). The buoys were deployed in December and retrieved 10 days later in January with a motorboat. The location was chosen due to its openness to long waves from the northern Baltic proper and its gently sloping bottom, which keeps the spatial wave height gradient small.

The mooring consisted of a 60 m rope from the anchor to the surface float and a 20 m rope from the float to the wave buoy. Unfortunately, LP3 started leaking during the field campaign, causing the MEMS sensor not to register data. We exclude this instrument from further analysis. After

retrieval, the instrument was cleaned and dried, and subsequent tests showed that no permanent harm was done to the electronics.

The mean wind speed during the measurement campaign was 4.1 m s^{-1} , but gusts at the Pakri weather station reached 16.5 m s^{-1} . During moderate wind speeds (over 6 m s^{-1}), the wind was mainly blowing from north or south. The maximum significant wave height was 1.22 m, and the maximum wave period was 5.97 s (Fig. 12). With the exception of the northerly winds, the waves and wind were misaligned. The

TABLE 5. Validation statistics. $N = 7486$ for H_{m0} . For other parameters, only cases where the DWR measured $H_{m0} > 0.25 \text{ m}$ were included ($N = 4827$).

Parameter	Bias	RMSD	SI (%)	Slope	R
H_{m0} (m)	0.01	0.04	8	0.99	0.99
T_{m02} (s)	0.06	0.10	3	1.02	0.98
T_{m01} (s)	0.08	0.12	3	1.02	0.98
T_{m10} (s)	0.16	0.23	5	1.04	0.96
T_c (s)	0.03	0.35	9	1.01	0.91
T_p (s)	0.03	0.89	21	0.99	0.66
κ^2 (–)	–0.02	0.04	23	0.85	0.92
σ_m (°)	3	5	9	1.09	0.88
σ_p (°)	2	8	25	1.03	0.49
	Bias	MAD			
θ_m (°)	1	7			
θ_p (°)	2	11			

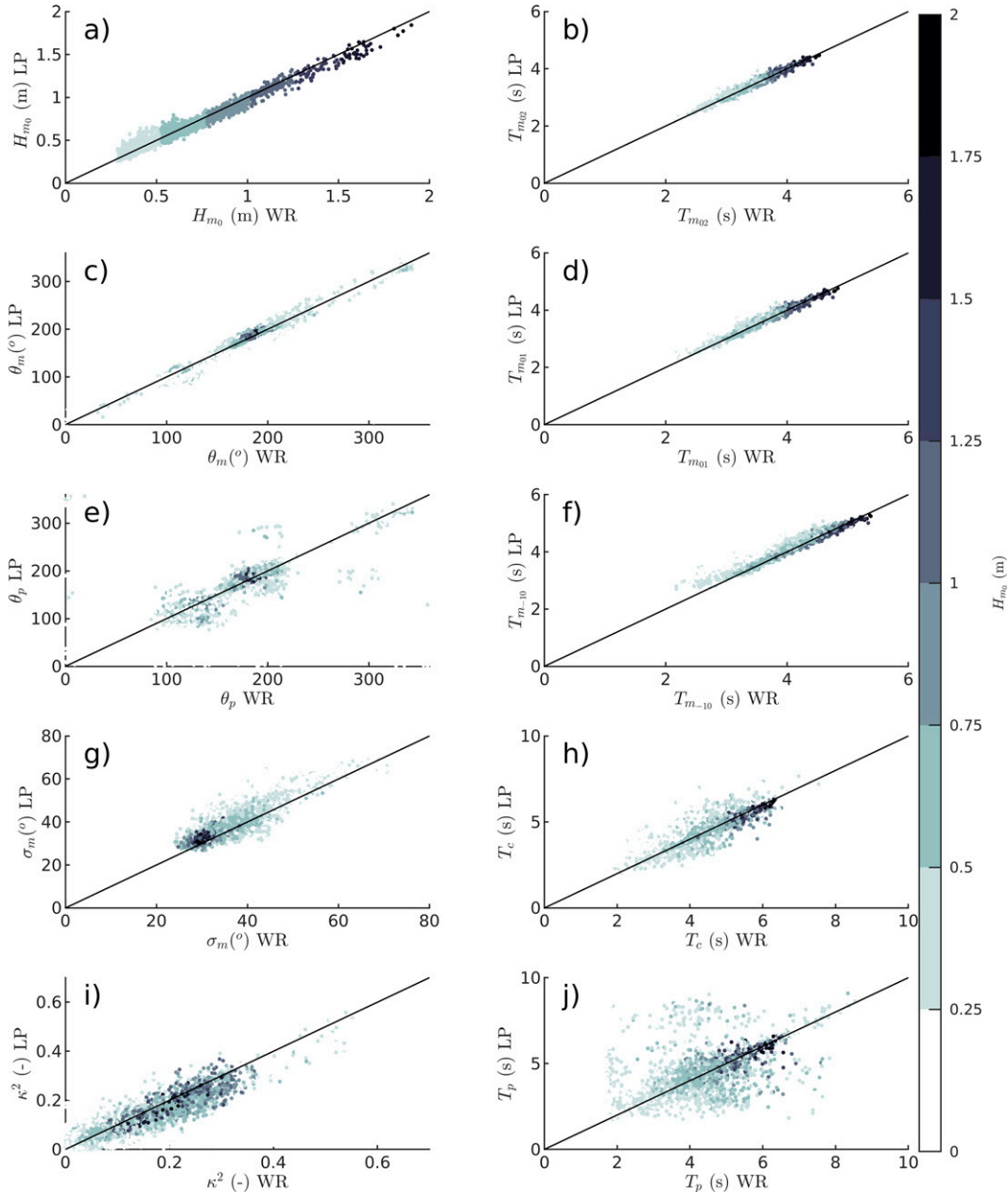


FIG. 10. Scatterplots of wave parameters during Suomenlinna validation deployment grouped by significant wave height. For the definition of variables see section 3.

misalignment was caused by the slanting fetch (Donelan et al. 1985; Petterson et al. 2010) and local topography. The slanting fetch also increased the directional spreading, since longer waves usually came from the west even though the short waves were aligned with the wind. For northerly winds, the wave and wind directions aligned, leading to a lower spreading compared to other wind directions.

The wave parameters between the different buoys agreed well (Fig. 12). The R^2 values (between LP1–LP2, LP1–LP4, and LP2–LP4) for significant wave height are over 0.99 and over 0.98 for mean wave period and for directional spreading.

The scatter index between H_{m0} was 3%–5%, reflecting the variability caused by sampling a random process (Donelan and Pierson 1983; Forristall et al. 1996).

The outlier was the mean wave direction of LP1. During the Suomenlinna UAV deployment (described in the next section) the release system malfunctioned and the buoy fell to rocky ground from approximately 2 m. Although bulk omnidirectional parameters and directional spread (which does not depend on true north reference) were reasonable, the yaw angle probably lost its north referencing capabilities, thus rendering the directional estimates unreliable.

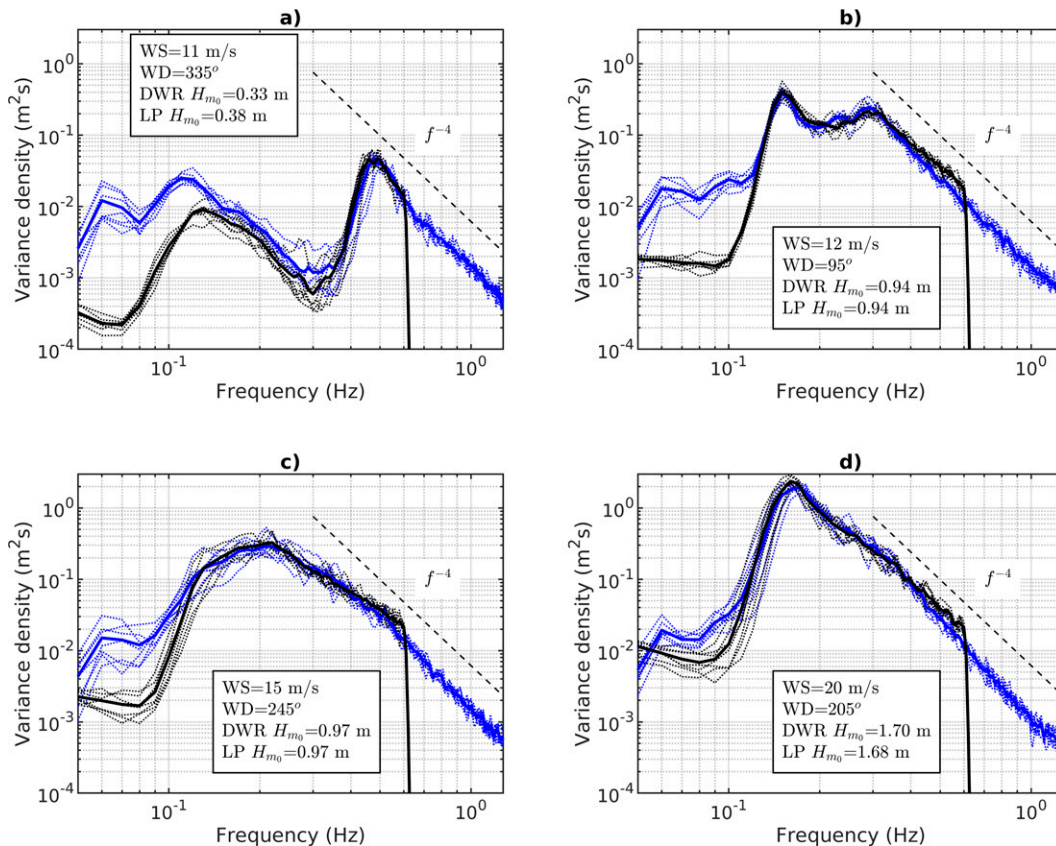


FIG. 11. Comparison of wave spectra at Suomenlinna for four different wind directions. Black lines are DWR; blue lines are LP. The thin dotted lines correspond to 30 min spectra, while the thick lines are averaged spectra of seven individual spectra.

c. Deployment with an unmanned aerial vehicle as a drifter

Vääna-Jõesuu (VJ) was the first ever deployment of LP with a UAV (Fig. 13a; Table 6). The buoy took about 2 h to drift back to the shore. Significant wave height was about 1 m, which also matched the value from a Copernicus CMEMS product (not shown here).

During the Suomenlinna deployment the buoy was dropped near the moored DWR (Fig. 13b). Wind gusts reached 15 m s^{-1} and the significant wave height was around 1 m with a wave direction from SSW. The buoy drifted until the next day and was picked up from a rocky island with a pilot boat.

At Leppneeme (Fig. 13c), the buoy was deployed from the shore. Although the wave direction was toward the shore, alongshore currents transported the buoy eastward for about 12 h before it got stuck in pancake ice close to the coast (Fig. 1c). The significant wave height was roughly 0.5 m.

The UAV ground stations were about 2 km from the deployment sites. During these three experiments, we operationally estimated the drift of the buoys with Seatrack web (<https://stw.smhi.se/>) and found that the wind leeway coefficient is around 2.5%, which is in the same range as similar spherical drifting buoys (Sutherland et al. 2020).

In the Gulf of Riga, we deployed three buoys from ice onto ice (Fig. 13d). UAV ground stations were approximately 2 km from shore at locations where pack ice was still safe to walk on. The buoys drifted with ice (with occasionally entering open water) up to month. One buoy stopped sending data 10 days after the deployment because it was brought directly from another (moored) experiment and the battery level was already low. Luckily, locals found it on shore and notified us using the contact information on the buoy. The results from the measurements in ice are described in section 5d.

The description of the UAV deployment procedures are presented in appendix D.

d. Waves in ice measurements

We deployed three drifting buoys in the eastern part of the Gulf of Riga (see section 5c). On 6 March 2021 a strong south-westerly wind event occurred, with the average measured wind speed and wind gusts being 15 and 21 m s^{-1} , respectively (Fig. 15a). Sentinel-1 overflights in the morning (0400 UTC) and in the evening (1600 UTC) showed that the ice edge retreated more than 10 km in about 12 h (Figs. 14a,b). The Copernicus CMEMS wave field forecast predicted a 2 m significant wave height near the ice edge close to the wave buoys

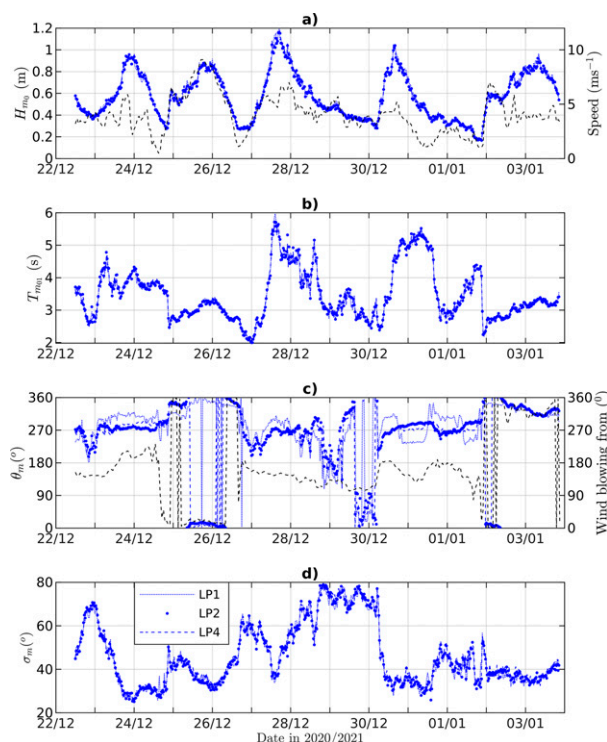


FIG. 12. Intercomparison of three LP's moored simultaneously about 100 m apart. The dashed black line marks (a) the wind speed from Pakri weather station and (c) the wind direction.

(Fig. 14c). The predicted peak period was 7 s, and the wave direction was between SW and W.

All three buoys started picking up the wave signal at around noon, and in the evening the significant wave height reached up to 1.2 cm at the buoy that was closest to the ice edge (Fig. 15b). The measured energy was above the sensor noise threshold for frequencies 0.10–0.14 Hz, and the surface displacement time series showed identifiable wave groups during the time of the maximum significant wave height (Figs. 15c,d). The roll and pitch angles deviated less than 1°, indicating that the buoys were firmly lodged in ice during the wave event.

6. Discussion

Research and development of prototype miniature wave buoys and loggers has increased in the last decade (e.g., Loeher et al. 2013; Kennedy et al. 2014; Hirakawa et al. 2016; Centurioni et al. 2017; Yurovsky and Dulov 2017; Farber et al. 2018; Skinner et al. 2018; Zong et al. 2019; Carandell et al. 2020; Cook et al. 2020). Small and affordable emerging wave buoys, like Spotter (Raghukumar et al. 2019) and LP, open up new possibilities for deploying large numbers of buoys simultaneously. With the Spotter, this has already been implemented on oceanic scales through assimilation of wave data in an operational wave model (Houghton et al. 2021; Smit et al. 2021). The ice resistance allows LP to be deployed in the marginal ice zone in large quantities, thus complementing the

existing waves-in-ice loggers (Kohout et al. 2015; Rabault et al. 2016; Montiel et al. 2018). This is of utmost importance since our knowledge on wave–ice interactions is still limited due to the low number of measurements made in ice (Squire 2020), especially measurements of the directional wave spectrum. Wave heights can be very low in the MIZ, and we have shown that a significant wave height of 1 cm is detectable by LP.

In the seasonally ice-covered water bodies, a cause for concern is the fact that operational devices have to be removed before the arrival of ice to avoid damaging the buoys. Nonetheless, the harshest wind conditions, and therefore wave conditions, can typically distort the wave statistics compiled purely from measurements (Björkqvist et al. 2018). Although LP is not designed to replace operational wave buoys, one possible application would be to complement operational measurements by replacing the operational buoy with LP when it is retrieved, thus capturing the open water time before freezing. LP should survive the ice, but if the buoy is lost, the financial implications are not as severe (buoy costs roughly EUR 5000) as a loss of an expensive operational buoy, nor do they threaten the continuity of the operational measurements as a whole.

Routine measurements also seldom capture waves shorter than 0.5–0.6 Hz, but LP extends the measurement range to 1.28 Hz (roughly 1 m wavelength). These shorter waves have been extensively studied with specialized research setups, e.g., using wave staffs or polarimetric cameras. The reason for the persistent scientific interest for this part of the wave regime is its decisive impact on many sea–atmosphere processes. Waves between 0.58 and 1.28 Hz have been shown to contribute 26% of the so-called Stokes drift (Lenain and Pizzo 2020), and they also carry a significant part of the wave-induced stress (Janssen 1991; Mueller and Veron 2009). These processes then affect the drift of the object, oil, and other surfactants and enhance the upper-layer turbulence and mixing, thus contributing to the sea–atmosphere fluxes of, e.g., CO₂. One possible niche for LP could be regions like archipelagos, fjords, or small lakes. In these areas, the short waves can carry a major part of the total wave energy, but establishing elaborate research wave measurement stations might not be feasible. From a practical perspective, the measurement of the high-frequency part might require cleaning of the buoy after some time to avoid accumulation of added mass and change of the buoy dimensions due to biofouling (Thomson et al. 2015; Campos et al. 2021).

Peak wave periods can reach 25 s in the world oceans (Hanafin et al. 2012), but low-frequency noise in wave measurements are unfortunately a common feature. Ashton and Johanning (2015) found spurious low-frequency energy caused by high drift forces brought about by currents and mooring. Low-frequency artifacts can also be created by a loss of the GPS signal in buoys that use the Doppler shift to measure the buoy velocity, e.g., the DWR-G4 (Björkqvist et al. 2016). We found that the MEMS sensor of the LP also suffers from low-frequency noise. This noise comes from the gyroscope random noise, which affects the pitch, roll, and yaw angles. These angles are then used to calculate the free acceleration inside the sensor, and the noise is further amplified by

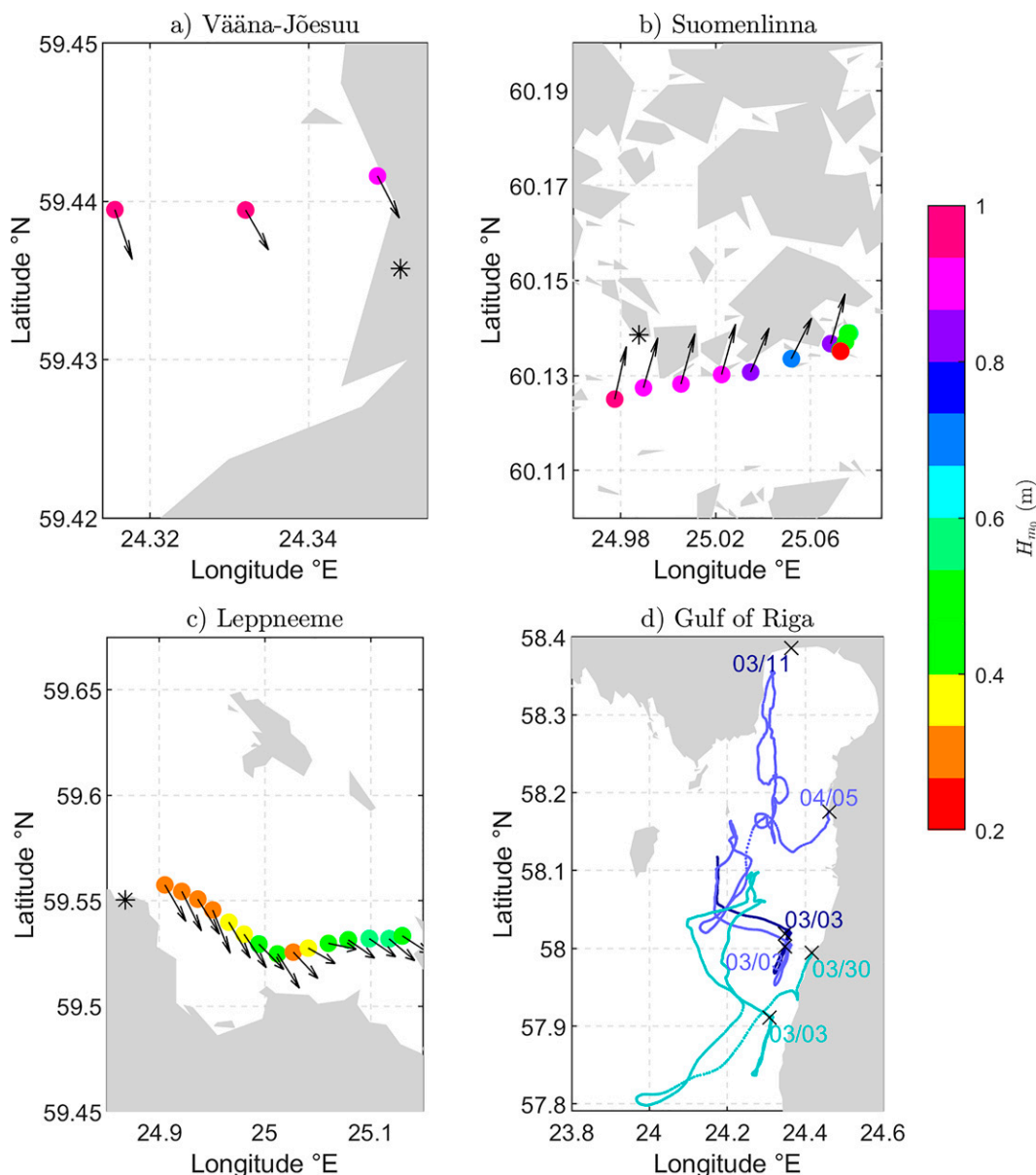


FIG. 13. (a)–(d) Drifting experiments results in four locations. UAV ground stations are marked with asterisks. In (d) only tracks of buoys are displayed, because most of the time no wave motion in ice was present. Also in (d), $\times$ marks the deployment or retrieval locations. For the dates of the experiments, see Table 6.

double integration. Low-frequency noise was also present in drifting buoys.

Yurovsky and Dulov (2020) also identified low-frequency noise, but did not present any correction method. Earle and Bush (1982) and Lang (1987) proposed a method for denoising by subtracting energy present at low frequencies from higher frequency. By comparing several months of LP measurements to coinciding Waverider measurements, we identified biases in the variance density both for the low and high frequencies, and concluded that these biases are possibly related. When the low-frequency part was denoised, we quantified the amount of variance that was lost and reintroduced

this variance back to the higher frequencies in proportion to the already existing variance (see appendix B). This correction was performed in Fourier space, which means that already the time series, determined through the inverse transform after integration in Fourier space was corrected. Although this method is partially ad hoc, it was shown to correct the final spectrum for both dominant waves and the spectral tail. Nonetheless, further development, not excluding changing the MEMS sensor, is currently pursued.

Nontypical deployments, such as aerial deployments by UAVs, have recently been used as an additional tool in observational studies (Zappa et al. 2020). UAVs are gaining

TABLE 6. An overview of drifting experiments where LainePoiss was deployed using a UAV. GoR refers to the Gulf of Riga.

	Deployment 1	Deployment 2	Deployment 3	Deployment 4	Deployment 5	Deployment 6
Device	LP1	LP1	LP5	LP2	LP5	LP6
Location	Vääna-Jõesuu (VJ)	Suomenlinna (SO)	Leppneeme	GoR	GoR	GoR
Type	Drifting	Drifting	Drifting	Drifting	Drifting	Drifting
Period	15 Aug 2019	6–7 Dec 2019	27–28 Feb 2021	3–11 Mar 2021	3 Mar–5 Apr	3–30 Mar
Mean U_{10} (m s^{-1})	8	13	8	In ice	In ice	In ice
Mean H_{m0} (m)	0.94	0.87	0.41	In ice	In ice	In ice
Mean T_{m10} (s)	4.2	4.4	2.7	In ice	In ice	In ice

popularity since they can carry lightweight sensors (Fuertes et al. 2019). These innovative solutions are partially motivated by environmental concerns, but they also allow to bypass the cost of a ship. Because of its low weight, LP can be deployed by an industrial UAV as a drifter. It has already been tested in deployments from shore and from sea ice with the deployment range extending to 2 km. These novel deployment techniques are still being established, but possible applications can include collecting data from surf zones or transporting a buoy onto the MIZ. UAV deployments will continue to be one part of the further development of LP.

It is possible to extend the operation time of LP by replacing rechargeable Li-ion batteries with lithium-thionyl chloride primary batteries. These batteries have an energy density that is about 2.5 times higher than that of rechargeable ones. This would extend the autonomous operating time to 5 months and make measurements in remote locations more feasible. Also, the buoy leaked on two occasions over the course of 18 field deployments. These incidents have been accounted for when designing the new hull (Fig. 16). The new hull is made of molded plastic and has a sealing O-ring on the top. External connectors for charging and accessing data have

been added to remove the need for opening the buoy and to enhance usability. The inertial sensor, electronics, and measurement methodology are kept the same. First laboratory tests of the new design on durability and waterproofness have been carried out successfully, but further testing is ongoing to validate this new design in the field.

7. Conclusions

A wave buoy [LainePoiss (LP)] was developed. LP weighs 3.5 kg, measures waves up to 1.28 Hz, has a rechargeable battery with 2 months of operation, and transmits wave parameters and spectra operationally over cellular or satellite networks. From our study, we can conclude the following:

- Wave parameters measured in the field were in good accordance with those measured by a nearby Directional Waverider: the bias of H_{m0} was 0.01 m, correlation 0.99, and scatter index of 8%. The bias of T_{m-10} was 0.14 s, correlation 0.98, and scatter index 4%. The mean absolute deviation of mean wave direction was 7°.
- The high-frequency part of the spectrum (up to 1.28 Hz) compared well to a wave gauge in the wave tank. The wave

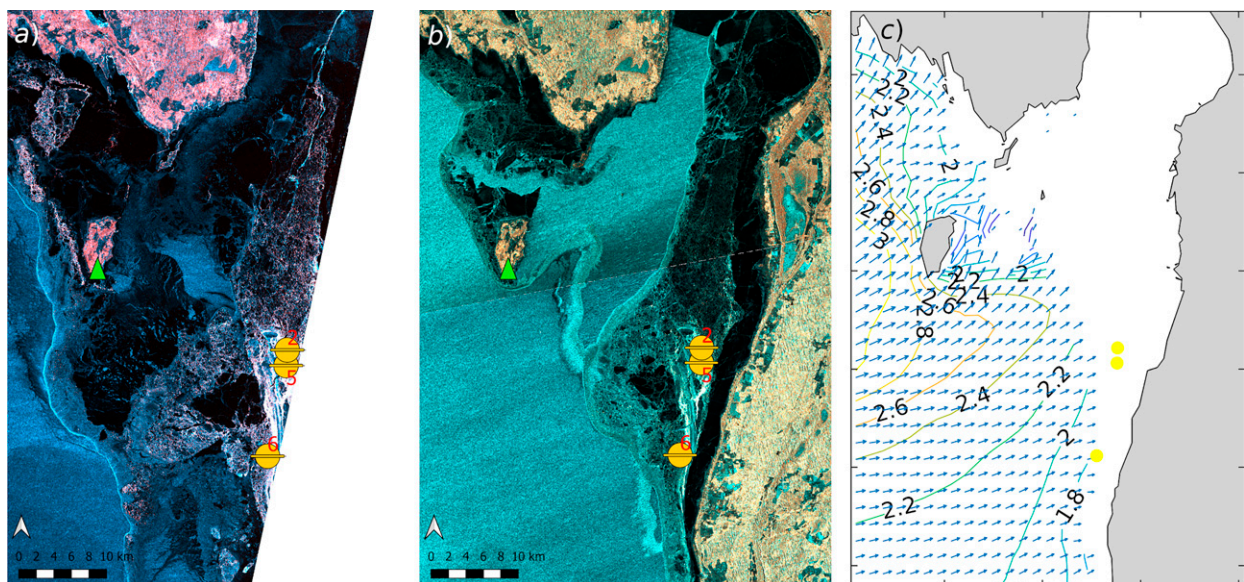


FIG. 14. (a) SAR image at 0400 UTC; (b) SAR image at 1600 UTC; (c) CMEMS model wave field (significant wave height, m; wave directions, arrows) at 1600 UTC. The locations of different LP's at 1600 UTC are shown with yellow marks. Kihnu weather station is marked with a green filled triangle. All panels are on 6 Mar 2021.

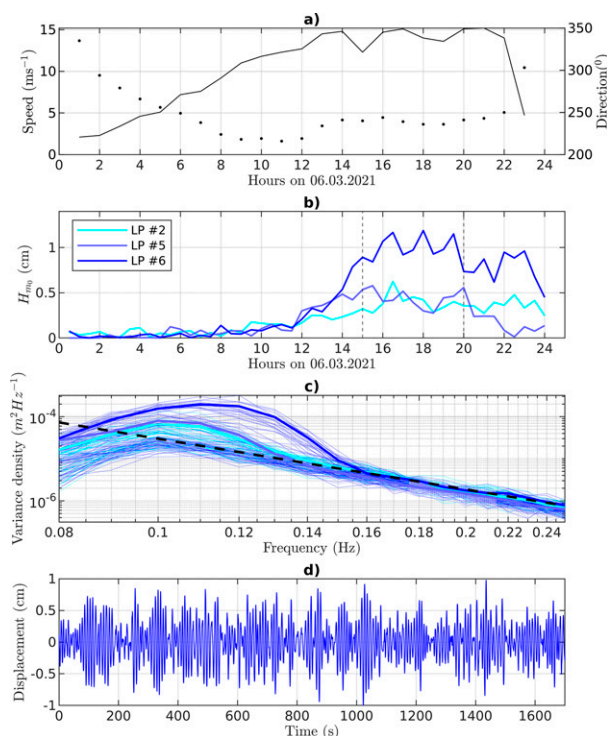


FIG. 15. Waves in ice measurements. (a) Wind speed and direction at Kihnu weather station (for location, see Fig. 14; direction marked with dots); (b) significant wave height from three buoys in ice; (c) spectra of three buoys with thin dotted lines corresponding to 30 min single spectra and thick lines corresponding to averaged spectra; dashed black line shows the sensor noise floor; (d) surface displacement for buoy 6 in ice.

spectrum of LP was validated up to a Waverider cutoff frequency of 0.58 Hz in the field, and the spectral tail of LP had an expected power-law behavior up to 1.28 Hz.

- LP can measure waves in ice with a significant wave height as low as 1 cm. This was confirmed in an over a month-long experiment where several LP's were deployed in ice in the Baltic Sea.
- Because of its low weight, LP can be deployed as a drifter to a distance of at least 2 km using an unmanned aerial vehicle (UAV).

Noise in the gyroscopic sensor resulted in low-frequency artifacts, but the displacement time series can be corrected (see appendix B). However, with the current validation and testing we can only confirm the buoy's capability to measure waves down to 0.1 Hz—this was a suitable range for the data we had available for the validation, although it precludes most of the oceanic applications. We will continue testing the wave buoy and expand our measurement areas outside of the Baltic Sea to capture longer waves. Depending on the MEMS sensor noise levels in these conditions, we can decide whether to replace the wave buoy's sensor for one with several times less noise.

Acknowledgments. This work was supported by the Personal Research Funding of the Estonian Ministry of Education and

Research (Grant PSG22). We thank Lauri Laakso and Roine Tuomo for helping with Suomenlinna field experiments. We appreciate discussions with Kimmo Kahma. We thank Rivo Uiboupin, Jüri Elken, Aarne Männik, and Sigrid Aas for keeping the administrative matters under control. We thank Tarmo Kõuts for helping us with the first versions of real-time display of data on a website. The first author thanks (little) Aran Alari for inspiring the wave buoy name LP. We thank the following institutions and enterprises for their successful participation in the applications of LainePoiss: the Maritime Administration of Estonia, the Estonian Environment Agency, AS Saarte Liinid, OÜ Lainemudel, TalTech Small Craft Competence Centre, and the City of Helsinki. We thank the anonymous reviewers for their constructive critique and comments—they helped us improve our article. Valdur Kaldvee and Kristjan Mölder are the founders, and the sole owners and employees of WiseParker OÜ, who produce the LainePoiss wave buoy presented in this manuscript. The development of LainePoiss has been made in cooperation with Tallinn University of Technology

Data availability statement. The data used in this manuscript will be made openly available upon the publication of the final paper. The LainePoiss MATLAB processing scripts are publicly available at <https://github.com/bjorkqvi/LPP/releases/tag/v1.0.0>.

APPENDIX A

The MTi-3 Sensor

This appendix provides the details of the Xsens MTi-3 sensor to the degree the information is publicly available. The sensor measures all 6 degrees of freedom (using a 3D accelerometer and 3D gyroscope) and provides the orientation (using a magnetometer). The technical specifications of the sensor (as provided by the manufacturer) can be found in Tables A1 and A2. The integrated microcontroller unit (MCU) synchronizes the various sensors, which is required for the onboard conversion of the acceleration signals from device to Earth-referenced frame. The role of the MCU is described by the manufacturer (Xsens 2019) as follows:

The MCU applies calibration models (unique to each sensor and including orientation, gain and bias offsets, plus more advanced relationships such as nonlinear temperature effects and other higher order terms) and runs the Xsens optimized strapdown algorithm, which performs highrate dead-reckoning calculations at 800 Hz, allowing accurate capture of high frequency motions and coning and sculling compensation. The Xsens sensor fusion engine combines all sensor inputs and optimally estimates the orientation, position and velocity at an output data rate of up to 100 Hz. The MTi 1-s is easily configurable for the outputs and depending on the application's needs can be set to use one of the filter profiles available within the Xsens sensor fusion engine.

The exact details of the built-in fusion algorithm—which transforms the signal from the sensor references to the Earth referenced frame—are not made public by the manufacturer.

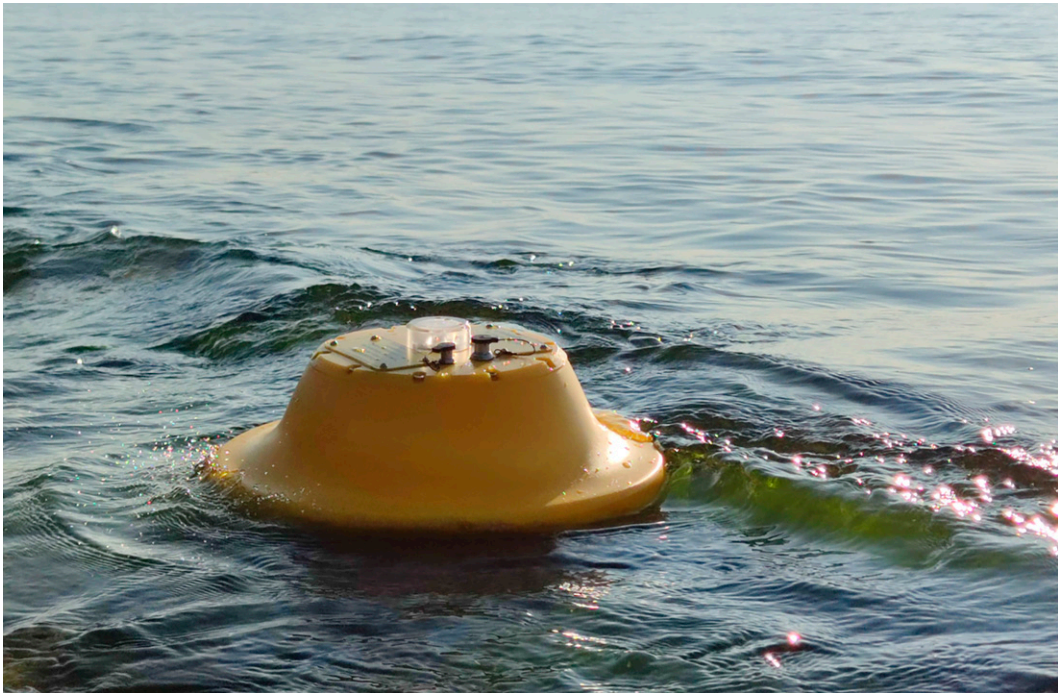


FIG. 16. New version of the buoy, with the charging and data acquisition connectors brought outside the buoy to reduce buoy openings.

However, the process includes an extended Kalman filter (Xsens 2015). The process and the different options for filtering are described by the manufacturer (Xsens 2019):

Xsens sensor fusion algorithm optimally estimates the orientation with respect to an Earth fixed frame utilizing the 3D inertial sensor data (orientation and velocity increments) and 3D magnetometer. The user can set the sensor fusion algorithm with different filter profiles in order to get the best performance based on the application scenario (see Table 16). These filter profiles contain predefined filter parameter settings suitable for different user application scenarios. In addition, all filter profiles can be used with the Active Heading Stabilization (AHS) setting, which significantly reduces heading drift during magnetic disturbances. The Inrun Compass Calibration (ICC) setting can be used to compensate for magnetic distortions that are caused by every object the MTi is attached to.

Of the available filter profiles, we used the profile “North referenced” (see Table 16 of Xsens 2019). We calibrated the magnetometer for hard iron distortion after assembling the whole device. This was done following the manufacturers specification using their Magnetic Field Mapper application. The process consisted of starting the application and rotating the buoy over a large number of orientations. After that, the tool calculated new calibration parameters and those were saved to the sensor memory. This had to be done only once after assembling the device. The calibration of the internal temperature sensor used for thermal compensation was already done by the manufacturer. The manufacturer states that the sensor has been calibrated to operate between -40° and 85°C .

To bring out the difference between the Earth referenced acceleration data and data in the body reference frame of the buoy, consider once more the benchmark tests from

TABLE A1. Specifications for the gyroscope and accelerometer of MTi-3 as given by the manufacturer (Xsens 2019).

	Gyroscope	Accelerometer
Standard full range	$\pm 2000^{\circ} \text{ s}^{-1}$	16g
In-run bias stability	$10^{\circ} \text{ h}^{-1}$	0.03 mg
Bandwidth (-3 dB)	255 Hz	324
Noise density	$0.007^{\circ} (\text{s } \sqrt{\text{Hz}})^{-1}$	$0.12 \text{ mg } (\sqrt{\text{Hz}})^{-1}$
Sensitivity variation	0.05%	0.05%
Nonlinearity	0.1%FS	0.5%FS
g sensitivity (calibrated)	$0.001^{\circ} (\text{s } g)^{-1}$	—
Max output frequency	800 Hz	800 Hz

TABLE A2. Specifications and performance statistics for the magnetometer and orientation of MTi-3 as given by the manufacturer (Xsens 2019).

	Value
Standard full range	8 G
Nonlinearity	0.2%
Total RMS noise	0.5 mG
Resolution	0.25 mG
Roll/pitch static RMS	0.5°
Roll/pitch dynamic RMS	0.8°
Yaw dynamic RMS	2°

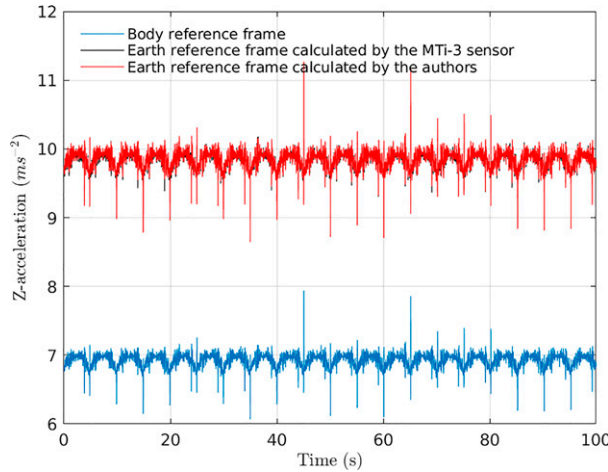


FIG. A1. Raw 50 Hz Z-oriented acceleration data in Earth reference frame and body reference frame.

section 4b. This time the sensor was placed so that the pitch angle was approximately 45° . We repeated the experiment for 5 s motion period (Fig. A1). The acceleration in the body reference frame is a fraction of the acceleration in the Earth reference frame, with a cosine dependence of roll and pitch angles (Bender et al. 2010). We calculated the Earth referenced acceleration by dividing the body referenced acceleration with the cosines of roll and pitch angles and found that it very well matches the values obtained with MTi-3 onboard processing (Fig. A1). The match of course is not one-to-one, since the MTi-3 fusion is much more elaborate and uses the accelerations from all three axes along with pitch and roll. For our benchmark test the implication of using Earth referenced data is straightforward. When the Earth referenced accelerations are used, calculated amplitude is off by 1 mm from the prescribed amplitude, while in the case of using acceleration data in body reference frame the calculated amplitude is 14 mm lower compared to the prescribed amplitude (for the test case of 5 s period motion).

APPENDIX B

Renormalization in Integration

The limitations of the sensor create low-frequency noise, but we have concluded that this noise is actually a misplaced signal from other frequencies. If the low-frequency noise is removed, it should be added back to the frequencies, so that the power spectrum fulfills the following:

$$E(f) = E_0(f) + \frac{\Delta\epsilon E_0(f)}{\sum_f E_0(f)} \quad (\text{B1})$$

$$= E_0(f) \left[1 + \frac{\Delta\epsilon}{\sum_f E_0(f)} \right], \quad (\text{B2})$$

where

$$\Delta\epsilon = \sum_f \hat{E}(f) - \sum_f E_0(f), \quad (\text{B3})$$

and $\hat{E}(f)$ and $E_0(f)$ are the power spectra of the unprocessed and denoised signals. $\Delta\epsilon$ quantifies how much variance density of the signal has been lost. Especially, if $\Delta\epsilon = 0$ then no further correction is necessary.

In practice, this denoising and compensation is performed during the integration, which results in a clean displacement signal from which the power spectrum can be calculated in a normal fashion without having to deal with the above compensation after the fact.

Low-frequency acceleration data are first denoised using the following procedure: the average squared Fourier amplitude is calculated for the frequencies 1/100 to 1/30 Hz:

$$\alpha = \left\langle \left| \hat{A}(f) \right|^2 \right\rangle_{1/100 \leq f \leq 1/30}, \quad (\text{B4})$$

where $\hat{A}(f) = \mathcal{F}[a(t)]$ is the Fourier transform of the acceleration signal and $|\hat{A}(f)|$ is its modulus.

This mean value (the noise) is then removed from the Fourier transform by

$$A_0(f) = \max \left\{ 0, \hat{A}(f) \frac{\hat{A}(f) - \alpha[1 - R_f(f)]}{\hat{A}(f)} \right\}, \quad (\text{B5})$$

where $R(f)$ is a response function:

$$R_f(f) = \begin{cases} 0, & 0 < f < f_1 \\ \frac{1}{2} \left[1 - \cos \left(\pi \frac{f - f_1}{f_2 - f_1} \right) \right], & f_1 \leq f \leq f_2 \\ 1, & f_2 < f \leq f_N \end{cases}. \quad (\text{B6})$$

Here $f_1 = 0.08$ Hz and $f_1 = 0.10$ Hz. In other words, the Fourier transform for $f > 0.10$ Hz is not touched. The 0.1 Hz cutoff was appropriate for our datasets in the Baltic Sea. For oceanic conditions this cutoff needs to be set to a lower value, but validating these lower frequencies would require additional field measurements that include longer waves.

Following, e.g., Rabault et al. (2020), the acceleration values are then integrated in Fourier space:

$$X_0(f) = -A_0(f)R_f(f) \cdot (2\pi f)^{-2}, \quad (\text{B7})$$

$$\hat{X}(f) = -\hat{A}(f)R_f(f) \cdot (2\pi f)^{-2}, \quad (\text{B8})$$

where $f_1 = 0.05$ Hz and $f_2 = 0.06$ Hz in the response function $R_f(f)$.

That is, $\hat{X}(f)$ and $X_0(f)$ are now the uncorrected and denoised Fourier transform of the displacement signal. We quantify how much of the squared amplitudes of the signal were lost by the low-frequency correction:

$$\Delta X^2 = \sum_f |\hat{X}(f)|^2 - \sum_f |X_0(f)|^2. \quad (\text{B9})$$

This lost signal information will be added back in a way that will compensate the power spectrum as outlined in Eq. (B2):

$$X(f) = X_0(f) \sqrt{1 + \frac{\Delta X^2}{\sum_f |X_0(f)|^2}} \quad (\text{B10})$$

$$= X_0(f) \sqrt{1 + \frac{\sum_f |\hat{X}(f)|^2 - \sum_f |X_0(f)|^2}{\sum_f |X_0(f)|^2}} \quad (\text{B11})$$

$$= X_0(f) \sqrt{1 + \frac{\sum_f |\hat{X}(f)|^2}{\sum_f |X_0(f)|^2} - 1} \quad (\text{B12})$$

$$= X_0(f) \sqrt{\frac{\sum_f |\hat{X}(f)|^2}{\sum_f |\hat{X}(f)|^2}}. \quad (\text{B13})$$

The displacement time series is then given by the inverse Fourier transform:

$$x(t) = \mathcal{F}^{-1}[X(f)]. \quad (\text{B14})$$

This time series is now corrected for the low-frequency artifacts, which has been redistributed to the other frequencies. The power spectrum or any other statistics can now be calculated from this displacement time series without further corrections.

APPENDIX C

Statistical Parameters

With N is the number of data points we used the following statistical parameters:

$$\text{Mean} = \bar{x}_i = \frac{1}{N} \sum_i x_i. \quad (\text{C1})$$

For angular quantities, θ , the mean is defined as

$$\bar{\theta}_i = \arctan \left[\frac{\sin(\theta_i)}{\cos(\theta_i)} \right] \quad (\text{C2})$$

and the difference

$$|\theta_1 - \theta_2| = \min\{|\theta_1 - \theta_2|, 360^\circ - |\theta_1 - \theta_2|\}. \quad (\text{C3})$$

The covariance is

$$\text{Cov}(x_i, r_i) = \sqrt{\frac{1}{N-1} \sum_i (x_i - \bar{x}_i)(r_i - \bar{r}_i)}. \quad (\text{C4})$$

where x_i are observations and r_i are the reference values (typically from the Waverider).

The variance and standard deviation are

$$\text{Var}(x_i) = \text{Cov}(x_i, x_i), \quad (\text{C5})$$

$$\text{Std}(x_i) = \sqrt{\text{Var}(x_i)}. \quad (\text{C6})$$

For cross comparison of datasets, we defined the following parameters:

$$\text{Bias} = \bar{x}_i - \bar{r}_i. \quad (\text{C7})$$

The root-mean-square deviation, mean absolute deviation and scatter index (in percent) are defined:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_i (x_i - r_i)^2}, \quad (\text{C8})$$

$$\text{MAD} = \frac{\sum_i |x_i - r_i|}{N}, \quad (\text{C9})$$

$$\text{SI} = 100 \sqrt{\frac{\sum_i [(x_i - \bar{x}_i) - (r_i - \bar{r}_i)]^2}{\sum_i r_i^2}}. \quad (\text{C10})$$

The slope is defined as a least squares fit $x = Kr$

$$K = \frac{\sum_i x_i r_i}{\sum_i r_i^2}, \quad (\text{C11})$$

and the Pearson correlation coefficient is

$$R = \frac{\text{Cov}(x_i, r_i)}{\text{Std}(x_i) \text{Std}(r_i)}. \quad (\text{C12})$$

APPENDIX D

Experiences from UAV Deployments

In this appendix, we document our experiences from six events where an industrial UAV was used to deploy LP as a floating device. Our deployments were made in Estonia and Finland during 2019–21. The guidelines and regulations governing the use of UAV's change and differ from country to country, especially outside of the European Union. Make sure to get correct information and follow any local laws and ordinances when considering deploying LP with a UAV. Also note that the steps below do not guarantee a safe or successful deployment.

The UAV should be regularly maintained to be sure of the condition of the equipment; motors, blades, batteries, and the remote controller have to be working properly. Firmware and software should also be up to date. Notifications through, e.g., the remote controller or pilot

application should be taken seriously, and they need to be resolved before flight. Before deploying the actual device, we recommend practicing with, e.g., sandbags that have the same weight as LP.

A successful UAV deployment starts with a comprehensive preparation and plan for the flight and requires at least two persons. The first person flies the drone while the second person assists at lift-off and landing. We used a DJI Matrice 600 Pro industrial drone with a release mechanism (Fig. 1b). A 5-m-long fishing thread (we used Dyneema 0.3 mm with a breaking strength of 25 kg) was found suitable for connecting the UAV and the wave buoy. A longer thread could make releasing the LP more convenient as the UAV can be higher, thus having a stronger signal strength. Nonetheless, a shorter thread minimizes the motion of the buoy during flight. The lift-off is the riskiest part: the second person has to hold the thread downwind and low to the ground to minimize the risk of the thread being caught by the UAV motors. This has to be done while keeping a distance of at least a few meters from the UAV. For the safety of the assisting person, the pilot should guide the UAV upward for the first couple of meters as quickly as possible. The rest of the lift-off can happen at a slower pace until the wave buoy is up in the air. We used a flight height of 20–30 m when transporting the wave buoy to the drop-off location.

With fully charged batteries at lift-off, we deployed the buoy at a distance of up to 2 km. We started guiding the UAV back toward the ground station before the battery dropped below 50%. In our deployments we flew upwind with the heavy weight, thus returning to the ground station downwind. We found that the battery charge at landing was still at about 20%.

Below is a step-by-step checklist that we have used in our deployments:

- The pilot does a UAV precheck a day before flight as well as on the field. If applicable, follow the messages from the pilot application.
- The second person, crouching at least 2 m from the UAV, holds the thread under tension between the UAV and the person's hand, ready to give slack as the UAV is ascending.
- The pilot loudly notifies the other persons before activating motors and makes sure everybody is ready for lift-off.
- The pilot ascends the first 2 m as quickly as possible, and the second person gives slack to the thread as the UAV takes off.
- A second person must hold the thread under tension between the UAV and their hand until the LP is up in the air.
- Thread should be downwind, even though the UAV might deviate also downwind where the second person is holding the thread.

Recommendations for the pilot:

- Flight height between 20 and 30 m.
- Do not fly farther than 50% of the battery capacity; ideally fly upwind with payload and return with tailwind.

REFERENCES

- Ashton, I., and L. Johanning, 2015: On errors in low frequency wave measurements from wave buoys. *Ocean Eng.*, **95**, 11–22, <https://doi.org/10.1016/j.oceaneng.2014.11.033>.
- Battjes, J. A., and G. P. van Vledder, 1984: Verification of Kimura's theory for wave group statistics. *Proc. 19th Int. Conf. on Coastal Engineering*, New York, NY, ASCE, 642–648, <https://doi.org/10.1061/9780872624382.044>.
- Bender, L. C., N. L. Guinasso, J. N. Walpert, and S. D. Howden, 2010: A comparison of methods for determining significant wave heights—Applied to a 3-m discus buoy during Hurricane Katrina. *J. Atmos. Oceanic Technol.*, **27**, 1012–1028, <https://doi.org/10.1175/2010JTECH0724.1>.
- Bishop, C. T., and M. A. Donelan, 1987: Measuring waves with pressure transducers. *Coastal Eng.*, **11**, 309–328, [https://doi.org/10.1016/0378-3839\(87\)90031-7](https://doi.org/10.1016/0378-3839(87)90031-7).
- Björkqvist, J.-V., 2020: Waves in archipelagos. Ph.D. thesis, University of Helsinki, 58 pp.
- , H. Pettersson, L. Laakso, K. K. Kahma, H. Jokinen, and P. Kosloff, 2016: Removing low-frequency artefacts from Datawell DWR-G4 wave buoy measurements. *Geosci. Instrum. Methods Data Syst.*, **5**, 17–25, <https://doi.org/10.5194/gi-5-17-2016>.
- , I. Lukas, V. Alari, G. P. van Vledder, S. Hulst, H. Pettersson, A. Behrens, and A. Männik, 2018: Comparing a 41-year model hindcast with decades of wave measurements from the Baltic Sea. *Ocean Eng.*, **152**, 57–71, <https://doi.org/10.1016/j.oceaneng.2018.01.048>.
- , H. Pettersson, and K. K. Kahma, 2019: The wave spectrum in archipelagos. *Ocean Sci.*, **15**, 1469–1487, <https://doi.org/10.5194/os-15-1469-2019>.
- Campos, R. M., H. Islam, T. R. Ferreira, and C. Guedes Soares, 2021: Impact of heavy biofouling on a nearshore heave-pitch-roll wave buoy performance. *Appl. Ocean Res.*, **107**, 102500, <https://doi.org/10.1016/j.apor.2020.102500>.
- Carandell, M., D. M. Toma, J. P. Pinto, M. Gasulla, and J. del Río, 2020: Impact on the wave parameters estimation of a kinetic energy harvester embedded into a drifter. *Global Oceans 2020: Singapore–U.S. Gulf Coast*, Biloxi, MS, IEEE, <https://doi.org/10.1109/IEEECONF38699.2020.9389127>.
- Cartwright, D. E., and M. S. Longuet-Higgins, 1956: The statistical distribution of the maxima of a random function. *Proc. Roy. Soc. London*, **237A**, 212–232, <https://doi.org/10.1098/rspa.1956.0173>.
- Cavaleri, L., 1980: Wave measurement using pressure transducer. *Oceanol. Acta*, **3**, 339–346.
- Centurioni, L., L. Braasch, E. Di Lauro, P. Contestabile, F. De Leo, R. Casotti, L. Franco, and D. Vicinanza, 2017: A new strategic wave measurement station off Naples port main breakwater. *Coastal Eng. Proc.*, **1**, 36, <https://doi.org/10.9753/icce.v35.waves.36>.
- Christakos, K., J.-V. Björkqvist, L. Tuomi, B. R. Furevik, and Ø. Breivik, 2021: Modelling wave growth in narrow fetch geometries: The white-capping and wind input formulations. *Ocean Modell.*, **157**, 101730, <https://doi.org/10.1016/j.ocemod.2020.101730>.
- Cook, A., A. Babanin, D. Sgarioto, P. Graham, J. Mathew, A. Skvortsov, R. Manasseh, and D. Tothova, 2020: A UAV 'mobile buoy' for measuring surface waves. *22nd Australasian Fluid Mechanics Conf.*, Brisbane, Australia, University of Queensland, <https://doi.org/10.14264/8e7141e>.

- Donelan, M. A., and W. J. Pierson, 1983: The sampling variability of estimates of spectra of wind-generated gravity waves. *J. Geophys. Res.*, **88**, 4381–4392, <https://doi.org/10.1029/JC088iC07p04381>.
- , J. Hamilton, and W. H. Hui, 1985: Directional spectra of wind-generated waves. *Philos. Trans. Roy. Soc. London*, **315A**, 509–562, <https://doi.org/10.1098/rsta.1985.0054>.
- Earle, M., and K. Bush, 1982: Strapped-down accelerometer effects on NDBO wave measurements. *OCEANS 82*, Washington, DC, IEEE, 838–848, <https://doi.org/10.1109/OCEANS.1982.1151908>.
- Farber, S., H. Allaka, I. Klein, and M. Groper, 2018: Estimating sea state using a low cost buoy. *2018 IEEE Int. Conf. on the Science of Electrical Engineering in Israel*, Eilat, Israel, IEEE, <https://doi.org/10.1109/ICSEE.2018.8646100>.
- Forristall, G. Z., J. C. Heideman, I. M. Leggett, B. Roskam, and L. Vanderschuren, 1996: Effect of sampling variability on hindcast and measured wave heights. *J. Waterw. Port Coastal Ocean Eng.*, **122**, 216–225, [https://doi.org/10.1061/\(ASCE\)0733-950X\(1996\)122:5\(216\)](https://doi.org/10.1061/(ASCE)0733-950X(1996)122:5(216)).
- Fuertes, F. C., L. Wilhelm, and F. Porté-Agel, 2019: Multirotor UAV-based platform for the measurement of atmospheric turbulence: Validation and signature detection of tip vortices of wind turbine blades. *J. Atmos. Oceanic Technol.*, **36**, 941–955, <https://doi.org/10.1175/JTECH-D-17-0220.1>.
- Gordon, L., and A. Lohrmann, 2002: Near-shore Doppler current meter wave spectra. *Proc. Int. Symp. on Ocean Wave Measurement and Analysis*, San Francisco, CA, American Society of Civil Engineers, 33–43, [https://doi.org/10.1061/40604\(273\)4](https://doi.org/10.1061/40604(273)4).
- Graber, H. C., E. A. Terray, M. A. Donelan, W. M. Drennan, J. C. Van Leer, D. B. Peters, and J. C. V. Leer, 2000: ASIS—A new air-sea interaction spar buoy: Design and performance at sea. *J. Atmos. Oceanic Technol.*, **17**, 708–720, [https://doi.org/10.1175/1520-0426\(2000\)017<0708:AANASI>2.0.CO;2](https://doi.org/10.1175/1520-0426(2000)017<0708:AANASI>2.0.CO;2).
- Hanafin, J. A., and Coauthors, 2012: Phenomenal sea states and swell from a North Atlantic storm in February 2011: A comprehensive analysis. *Bull. Amer. Meteor. Soc.*, **93**, 1825–1832, <https://doi.org/10.1175/BAMS-D-11-00128.1>.
- Herbers, T. H. C., P. F. Jessen, T. T. Janssen, D. B. Colbert, and J. H. MacMahan, 2012: Observing ocean surface waves with GPS-tracked buoys. *J. Atmos. Oceanic Technol.*, **29**, 944–959, <https://doi.org/10.1175/JTECH-D-11-00128.1>.
- Hirakawa, Y., T. Takayama, T. Hirayama, and H. Susaki, 2016: Development of mini-buoy for short term measurement of ocean wave. *OCEANS 2016 MTS/IEEE Monterey*, Monterey, CA, IEEE, <https://doi.org/10.1109/OCEANS.2016.7761337>.
- Holthuijsen, L., 2007: *Waves in Oceanic and Coastal Waters*. Cambridge University Press, 387 pp.
- Houghton, I., P. Smit, D. Clark, C. Dunning, A. Fisher, N. Nidzieko, P. Chamberlain, and T. Janssen, 2021: Performance statistics of a real-time Pacific Ocean weather sensor network. *J. Atmos. Oceanic Technol.*, **38**, 1047–1058, <https://doi.org/10.1175/JTECH-D-20-0187.1>.
- Janssen, P. A., 1991: Quasi-linear theory of wind-wave generation applied to wave forecasting. *J. Phys. Oceanogr.*, **21**, 1631–1642, [https://doi.org/10.1175/1520-0485\(1991\)021<1631:QLTOWW>2.0.CO;2](https://doi.org/10.1175/1520-0485(1991)021<1631:QLTOWW>2.0.CO;2).
- Kennedy, D., M. Walsh, and B. O'Flynn, 2014: Development of a miniature, low-cost wave measurement solution. *2014 Oceans—St. John's*, St. John's, NL, Canada, IEEE, <https://doi.org/10.1109/OCEANS.2014.7003217>.
- Kitaigorodskii, S. A., 1983: On the theory of the equilibrium range in the spectrum of wind-generated gravity waves. *J. Phys. Oceanogr.*, **13**, 816–827, [https://doi.org/10.1175/1520-0485\(1983\)013<0816:OTTOTE>2.0.CO;2](https://doi.org/10.1175/1520-0485(1983)013<0816:OTTOTE>2.0.CO;2).
- Kodaira, T., T. Waseda, T. Nose, K. Sato, J. Inoue, J. Voermans, and A. Babanin, 2020: Observation of on-ice wind waves under grease ice in the western Arctic Ocean. *Polar Sci.*, **27**, 100567, <https://doi.org/10.1016/j.polar.2020.100567>.
- Kohout, A. L., B. Penrose, S. Penrose, and M. J. Williams, 2015: A device for measuring wave-induced motion of ice floes in the Antarctic marginal ice zone. *Ann. Glaciol.*, **56**, 415–424, <https://doi.org/10.3189/2015AoG69A600>.
- Lancaster, O., R. Cossu, S. Boulay, S. Hunter, and T. E. Baldock, 2021: Comparative wave measurements at a wave energy site with a recently developed low-cost wave buoy (Spotter), ADCP, and pressure loggers. *J. Atmos. Oceanic Technol.*, **38**, 1019–1033, <https://doi.org/10.1175/JTECH-D-20-0168.1>.
- Lang, N., 1987: The empirical determination of a noise function for NDBC buoys with strapped-down accelerometers. *OCEANS'87*, Halifax, NS, Canada, IEEE, 225–228, <https://doi.org/10.1109/OCEANS.1987.1160904>.
- Le Merle, E., D. Hauser, C. Peureux, L. Aouf, P. Schippers, C. Dufour, and A. Dalphiné, 2021: Directional and frequency spread of surface ocean waves from SWIM measurements. *J. Geophys. Res. Oceans*, **126**, e2021JC017220, <https://doi.org/10.1029/2021JC017220>.
- Lenain, L., and N. Pizzo, 2020: The contribution of high-frequency wind-generated surface waves to the Stokes drift. *J. Phys. Oceanogr.*, **50**, 3455–3465, <https://doi.org/10.1175/JPO-D-20-0116.1>.
- Lin, Z., T. A. A. Adcock, and M. L. McAllister, 2021: Estimating ocean wave directional spreading using wave following buoys: A comparison of experimental buoy and gauge data. *J. Ocean Eng. Mar. Energy*, **8**, 83–97, <https://doi.org/10.1007/s40722-021-00218-7>.
- Loehr, H., and Coauthors, 2013: Development of a low cost wave buoy by utilising smartphone technology. *21st Australasian Coastal and Ocean Engineering Conf.—14th Australasian Port and Harbour Conf.*, Barton, Australia, Engineers Australia.
- Longuet-Higgins, M. S., 1961: Observations of the directional spectrum of sea waves using the motions of a floating buoy. *Ocean Wave Spectra*, Prentice-Hall, 111–136.
- Montiel, F., V. Squire, M. Doble, J. Thomson, and P. Wadhams, 2018: Attenuation and directional spreading of ocean waves during a storm event in the autumn Beaufort Sea marginal ice zone. *J. Geophys. Res. Oceans*, **123**, 5912–5932, <https://doi.org/10.1029/2018JC013763>.
- Mueller, J. A., and F. Veron, 2009: Nonlinear formulation of the bulk surface stress over breaking waves: Feedback mechanisms from air-flow separation. *Bound.-Layer Meteor.*, **130**, 117–134, <https://doi.org/10.1007/s10546-008-9334-6>.
- Munk, W. H., 1950: Origin and generation of waves. *Proc. First Conf. Coastal Engineering*, Long Beach, CA, ASCE, 1–4, <https://doi.org/10.9753/icce.v1.1>.
- Pearman, D., T. Herbers, T. Janssen, H. van Ettinger, S. McIntyre, and P. Jessen, 2014: Drifter observations of the effects of shoals and tidal-currents on wave evolution in San Francisco Bight. *Cont. Shelf Res.*, **91**, 109–119, <https://doi.org/10.1016/j.csr.2014.08.011>.
- Petterson, H., H. C. Graber, D. Hauser, C. Quentin, K. K. Kahma, W. M. Drennan, and M. A. Donelan, 2003: Directional wave measurements from three wave sensors

- during the FETCH experiment. *J. Geophys. Res.*, **108**, 8061, <https://doi.org/10.1029/2001JC001164>.
- , K. K. Kahma, and L. Tuomi, 2010: Wave directions in a narrow bay. *J. Phys. Oceanogr.*, **40**, 155–169, <https://doi.org/10.1175/2009JPO4220.1>.
- Phillips, O. M., 1985: Spectral and statistical properties of the equilibrium range in wind-generated gravity waves. *J. Fluid Mech.*, **156**, 505–531, <https://doi.org/10.1017/S0022112085002221>.
- Rabault, J., G. Sutherland, B. Ward, K. H. Christensen, T. Halsne, and A. Jensen, 2016: Measurements of waves in landfast ice using inertial motion units. *IEEE Trans. Geosci. Remote Sens.*, **54**, 6399–6408, <https://doi.org/10.1109/TGRS.2016.2584182>.
- , —, O. Gundersen, A. Jensen, A. Marchenko, and Ø. Breivik, 2020: An open source, versatile, affordable waves in ice instrument for scientific measurements in the polar regions. *Cold Reg. Sci. Technol.*, **170**, 102955, <https://doi.org/10.1016/j.coldregions.2019.102955>.
- Raghukumar, K., G. Chang, F. Spada, C. Jones, T. Janssen, and A. Gans, 2019: Performance characteristics of “Spotter,” a newly developed real-time wave measurement buoy. *J. Atmos. Oceanic Technol.*, **36**, 1127–1141, <https://doi.org/10.1175/JTECH-D-18-0151.1>.
- Skinner, E., M. Rooney, and M. Hinders, 2018: Low-cost wave characterization modules for oil spill response. *J. Ocean Eng. Sci.*, **3**, 96–108, <https://doi.org/10.1016/j.joes.2018.05.003>.
- Smit, P. B., I. A. Houghton, K. Jordanova, T. Portwood, E. Shapiro, D. Clark, M. Sosa, and T. T. Janssen, 2021: Assimilation of significant wave height from distributed ocean wave sensors. *Ocean Modell.*, **159**, 101738, <https://doi.org/10.1016/j.oceanmod.2020.101738>.
- Squire, V. A., 2020: Ocean wave interactions with sea ice: A reappraisal. *Annu. Rev. Fluid Mech.*, **52**, 37–60, <https://doi.org/10.1146/annurev-fluid-010719-060301>.
- Sutherland, G., and Coauthors, 2020: Evaluating the leeway coefficient of ocean drifters using operational marine environmental prediction systems. *J. Atmos. Oceanic Technol.*, **37**, 1943–1954, <https://doi.org/10.1175/JTECH-D-20-0013.1>.
- Thomson, J., 2012: Wave breaking dissipation observed with “swift” drifters. *J. Atmos. Oceanic Technol.*, **29**, 1866–1882, <https://doi.org/10.1175/JTECH-D-12-00018.1>.
- , and Coauthors, 2015: Biofouling effects on the response of a wave measurement buoy in deep water. *J. Atmos. Oceanic Technol.*, **32**, 1281–1286, <https://doi.org/10.1175/JTECH-D-15-0029.1>.
- Veras Guimarães, P., and Coauthors, 2018: A surface kinematics buoy (SKIB) for wave–current interaction studies. *Ocean Sci.*, **14**, 1449–1460, <https://doi.org/10.5194/os-14-1449-2018>.
- Voermans, J., P. Smit, T. Janssen, and A. Babanin, 2020: Estimating wind speed and direction using wave spectra. *J. Geophys. Res. Oceans*, **125**, e2019JC015717, <https://doi.org/10.1029/2019JC015717>.
- Wadhams, P., 1978: Wave decay in the marginal ice zone measured from a submarine. *Deep-Sea Res.*, **25**, 23–40, [https://doi.org/10.1016/S0146-6291\(21\)00004-7](https://doi.org/10.1016/S0146-6291(21)00004-7).
- Xsens, 2015: Data sheet MTi 1-series: 3D AHRS/VRU/IMU module. Xsens Doc. MT0512P, Revision B, 35 pp., http://www.sensores-de-medida.es/uploads/xsens_mti-1-series_datasheet.pdf.
- , 2019: MTi 1-series datasheet: IMU, VRU, AHRS and GNSS/INS module. Xsens Doc. MT0512P, Revision 2019.A, 40 pp., <https://www.xsens.com/hubfs/Downloads/Manuals/MTi-1-series-datasheet.pdf>.
- Young, I. R., 1995: The determination of confidence limits associated with estimates of the spectral peak frequency. *Ocean Eng.*, **22**, 669–686, [https://doi.org/10.1016/0029-8018\(95\)00002-3](https://doi.org/10.1016/0029-8018(95)00002-3).
- Yurovsky, Y. Y., and V. Dulov, 2017: Compact low-cost Arduino-based buoy for sea surface wave measurements. *2017 Progress in Electromagnetics Research Symp.—Fall*, Singapore, IEEE, 2315–2322, <https://doi.org/10.1109/PIERS-FALL.2017.8293523>.
- , and —, 2020: MEMS-based wave buoy: Towards short wind-wave sensing. *Ocean Eng.*, **217**, 108043, <https://doi.org/10.1016/j.oceaneng.2020.108043>.
- Zappa, C. J., S. M. Brown, N. J. M. Laxague, T. Dhakal, R. A. Harris, A. M. Farber, and A. Subramaniam, 2020: Using ship-deployed high-endurance unmanned aerial vehicles for the study of ocean surface and atmospheric boundary layer processes. *Front. Mar. Sci.*, **6**, 777, <https://doi.org/10.3389/fmars.2019.00777>.
- Zong, Y., H.-Y. Cheng, H. Chien, and B. Koppe, 2019: Miniature wave buoy—Laboratory and field tests for development of a robust low-cost measuring technique. *Coastal Structures 2019*, Hannover, Germany, Bundesanstalt für Wasserbau, 453–462, https://doi.org/10.18451/978-3-939230-64-9_046.